Population analysis of Binary Black Holes

Yin-Jie Li （李银杰）, Purple Mountain Observatory

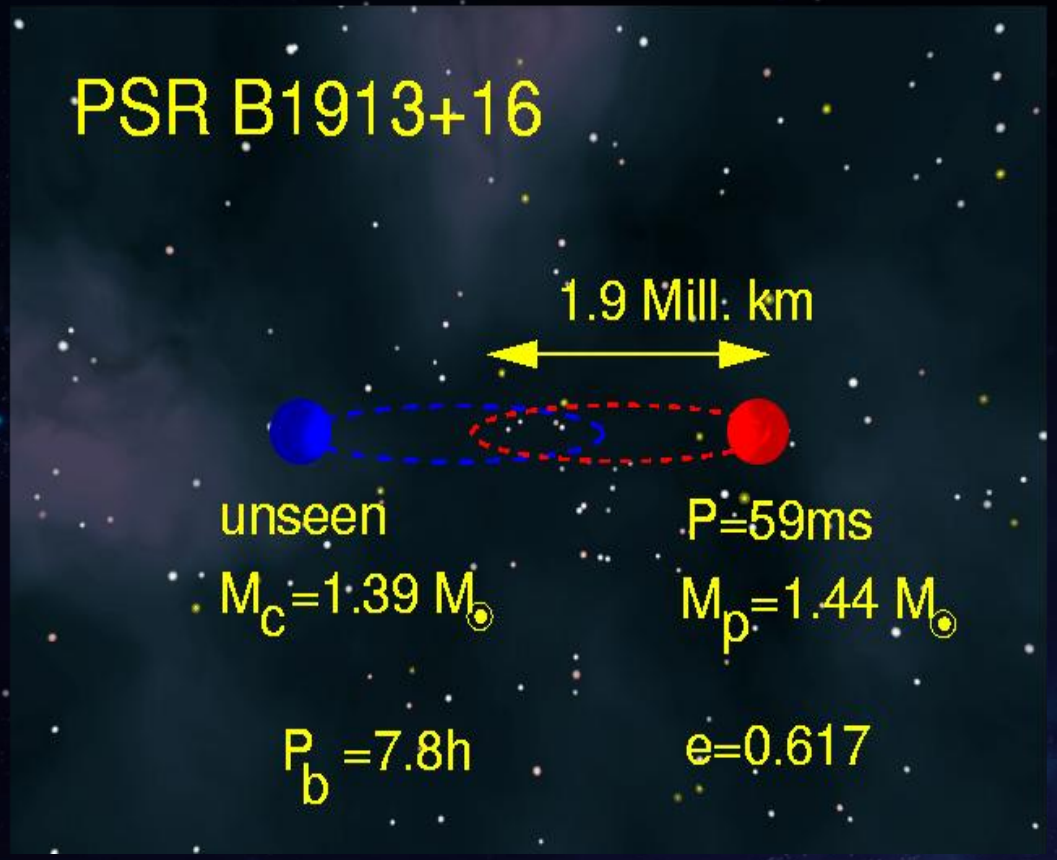
Outline

- Introduction：Formation channels of BBHs
- Signatures in GW observation
- Population analysis：Hierarchical Bayesian Inference

Formation channels of BBHs:

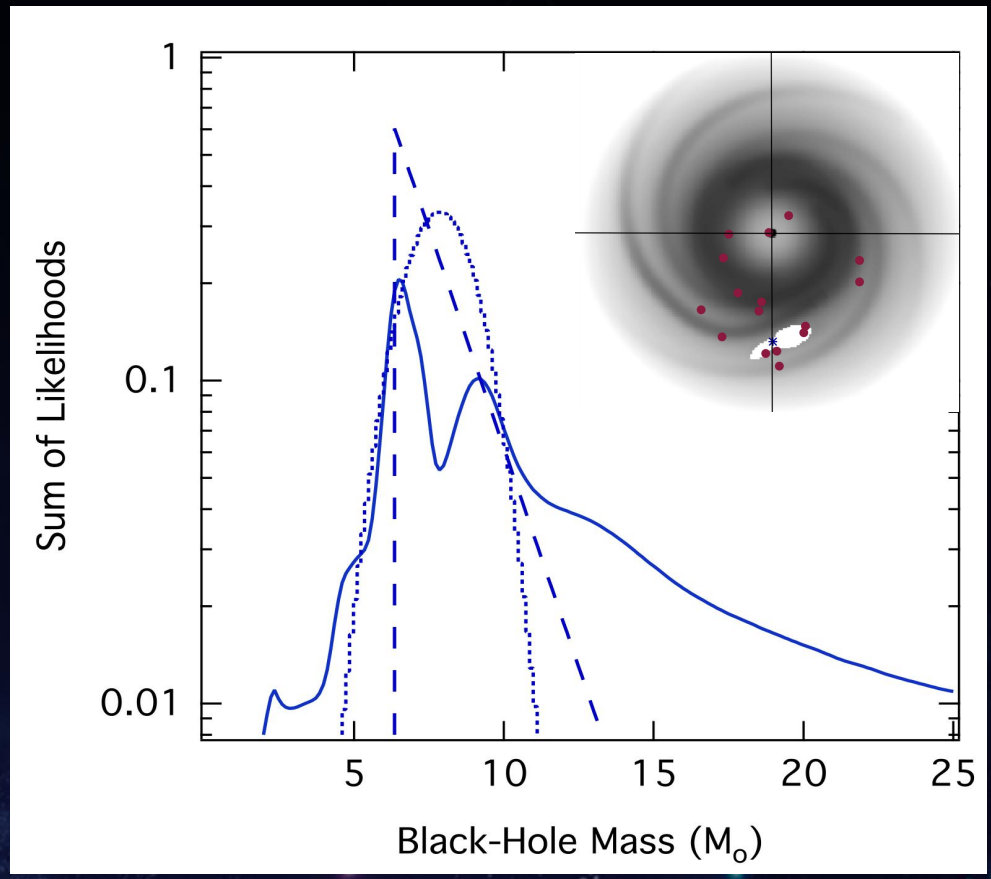
Why is GW150914 a BBH merger?
Why are they so massive?

neutron star binaries



Joseph Hooton Taylor Jr. with Russell A. Hulse won the Nobel Prize in Physics in 1993.

Mass of BHs in X-ray binaries

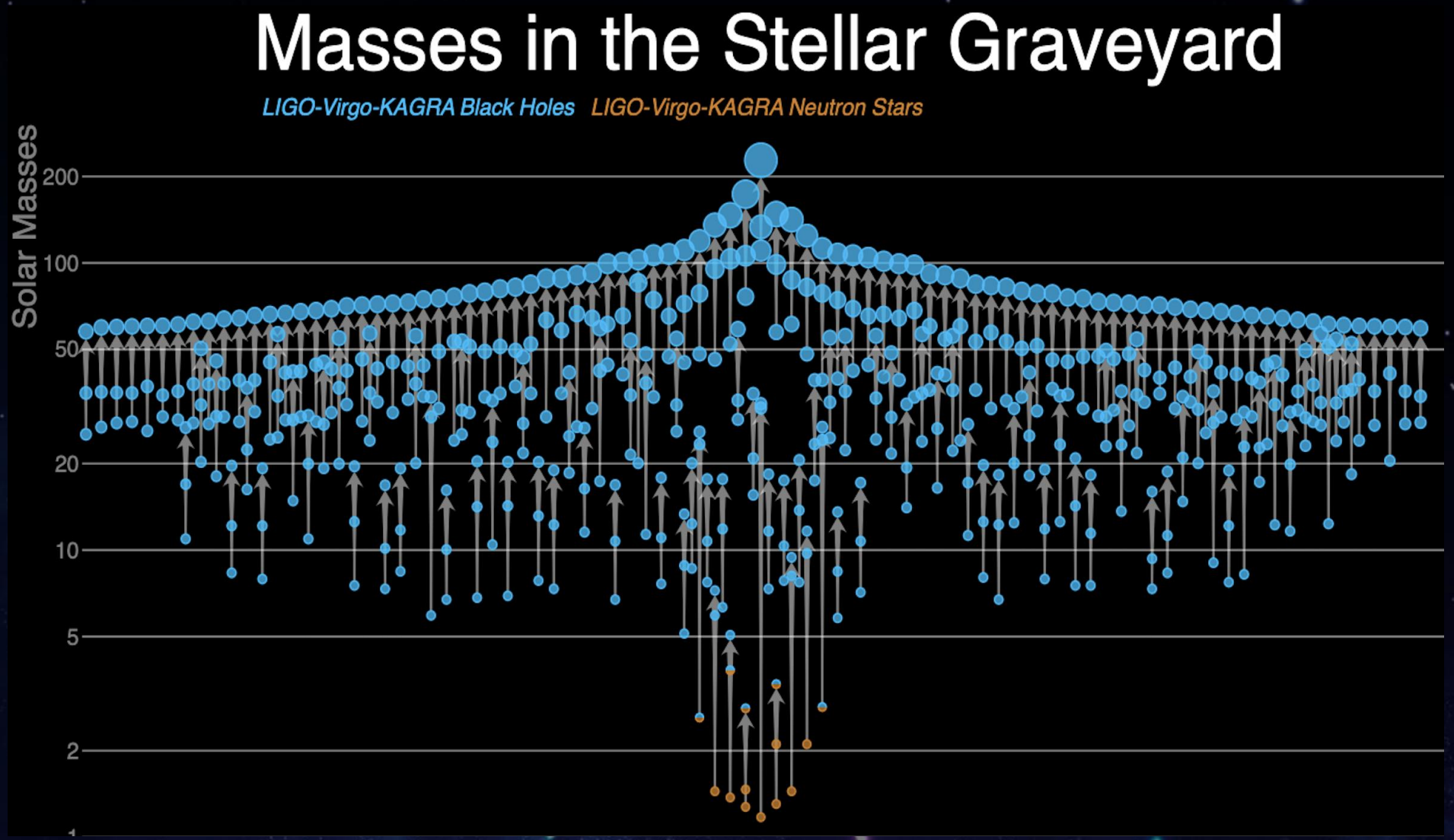


Ozel et al. 2010, ApJ
Black holes in our Galaxy

More than 300 events have been observed

Question:
How do they form and merge?

Where are they?

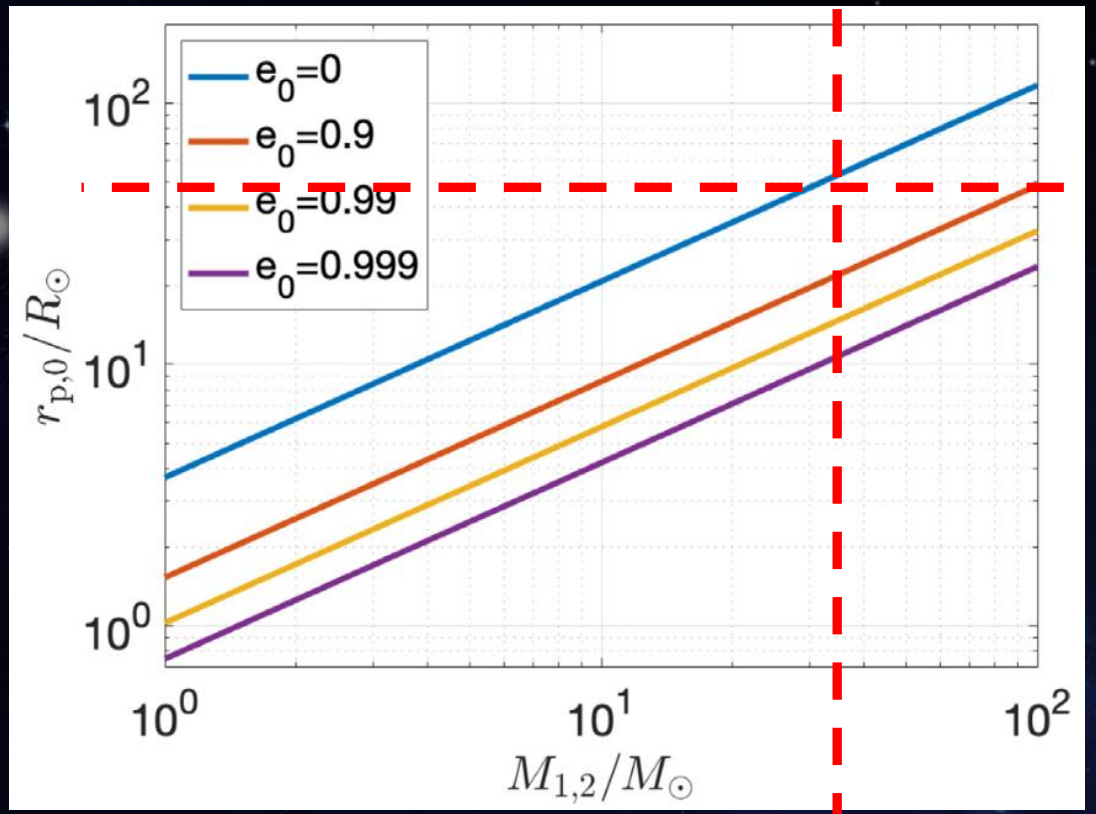


Most events are binary black holes

演化路径的切入点：

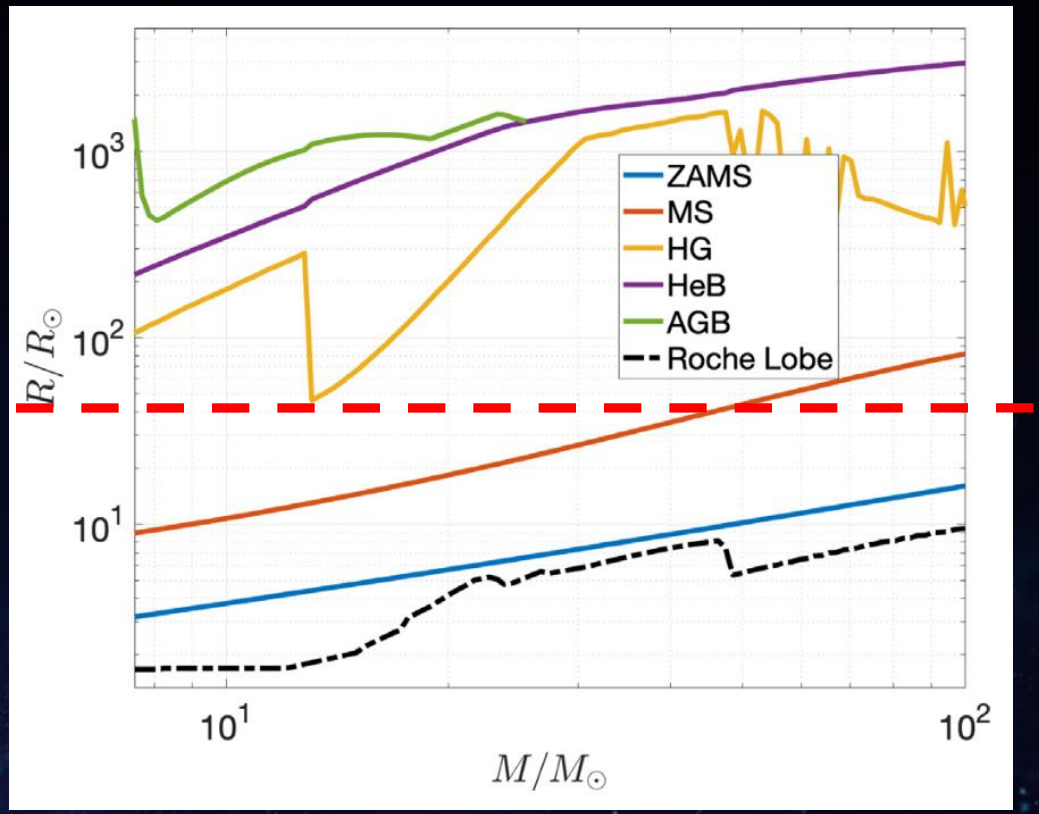
The separation problem

- Only very tight binaries can merge via gravitational waves.



GW150914

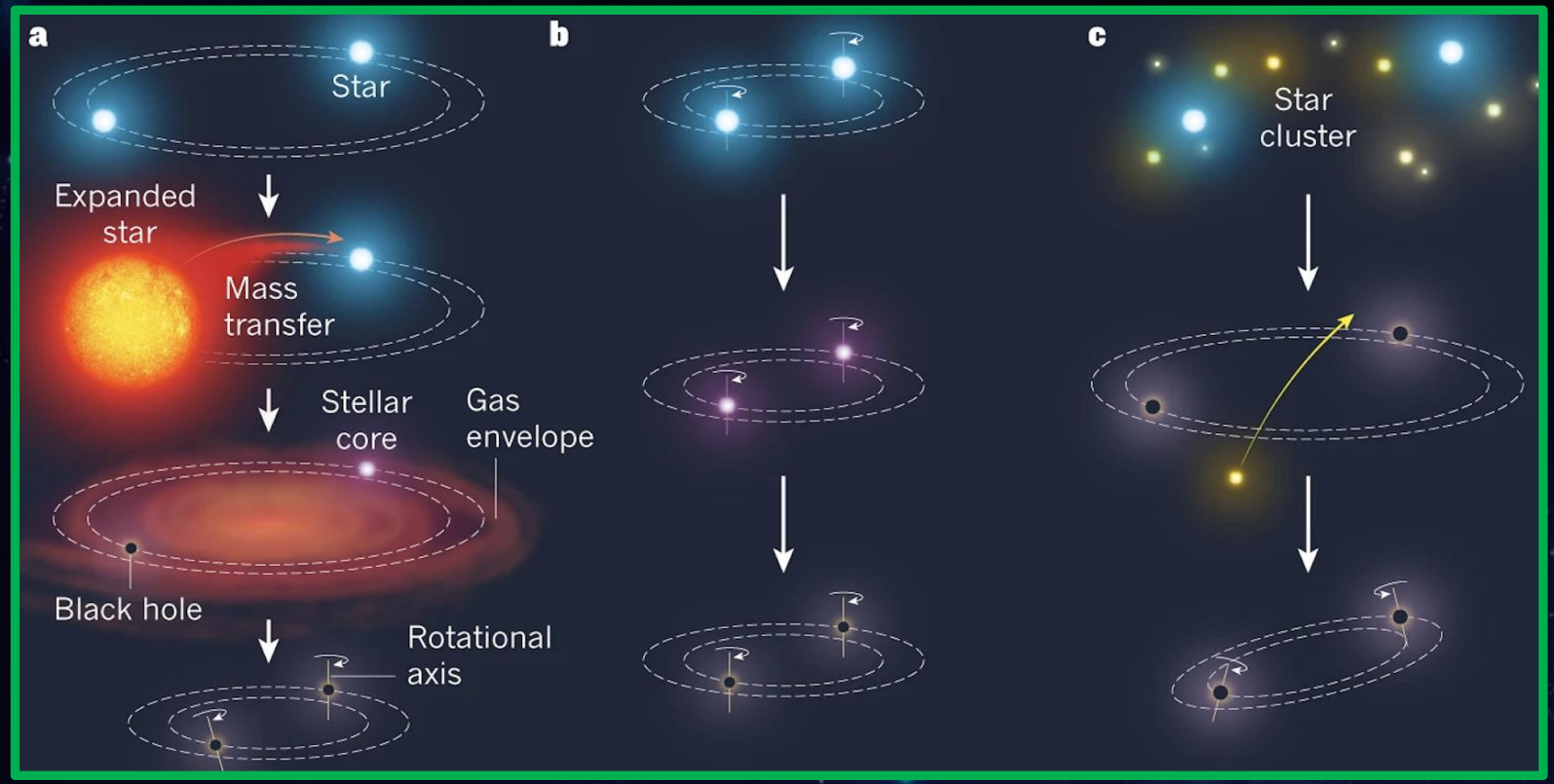
- But the black holes' parent stars cannot get so close

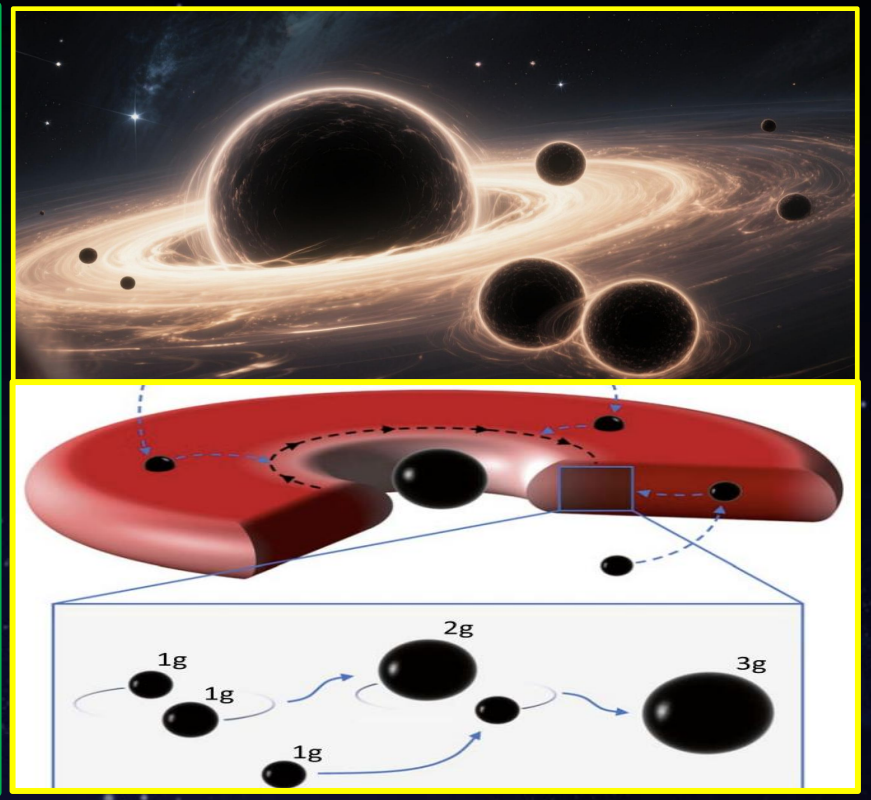


Mendal&Farmer 2022 Phys.Rept.

The canditate scenarios

- Coming closer later in life （common-envelope phase）
- Stars not expand （Population III stars）
- The black-hole matchmaking club: dynamical formation in dense stellar environments
- shrink orbits by gas (AGN disks)
- 。 。 。 。 。 。





Gayathri et al. 2017 ApJL

Three main formation channels (Evironments)

Isolated Binary Evolution



Common envelope

Stable mass transfer

Pop III stars

Chemically homogeneous evolution

Wide binaries + flybys

Secular evolution of triples/quadruples

Triples w/ common envelope

Globular Clusters

Nuclear clusters

Open clusters

Young massive clusters

Ultra-dwarf galaxy mergers

Active galactic nuclei

Dynamical Assembly



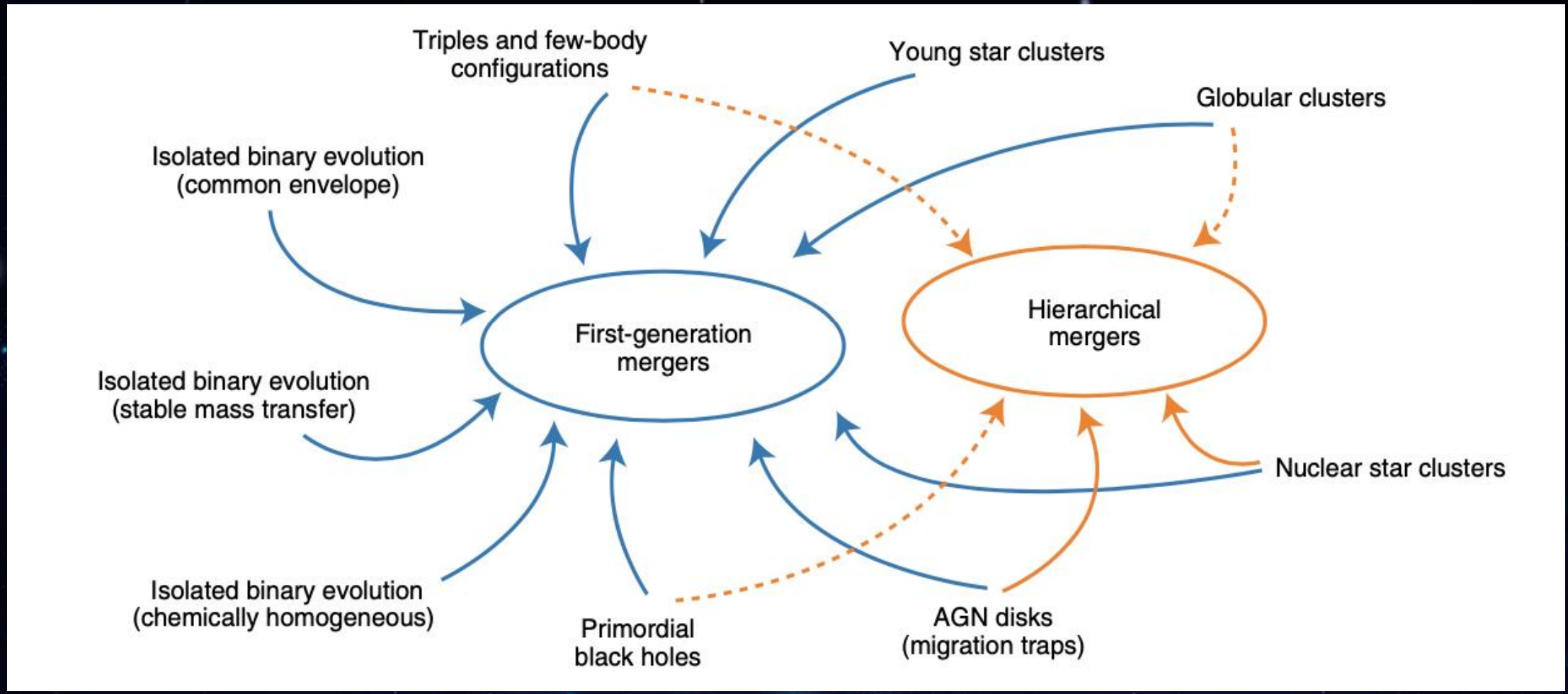
Gaseous Environments

Non-Astrophysical

Primordial black holes/ Dark matter

Keys: hierarchical mergers depend on Evironments

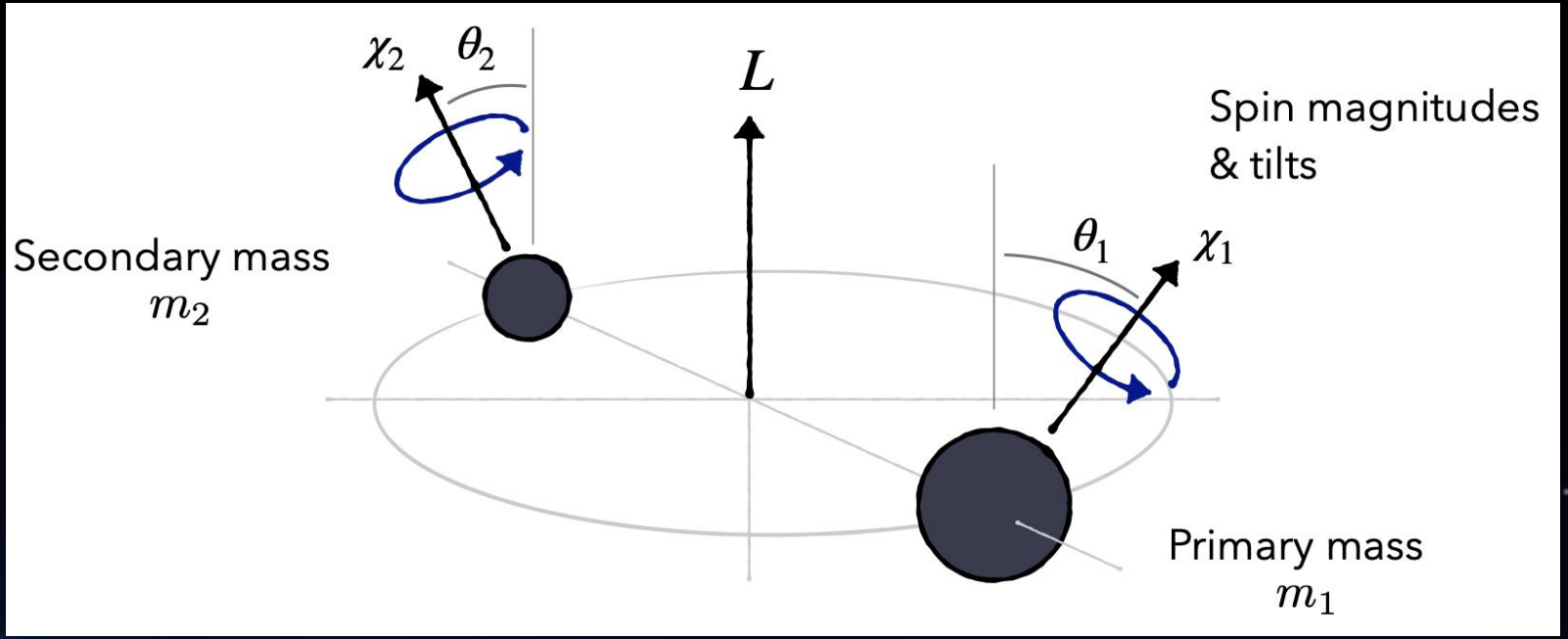
- isolated channels only generate first-generation mergers, whereas star clusters and AGN disk channels may generate hierarchial mergers

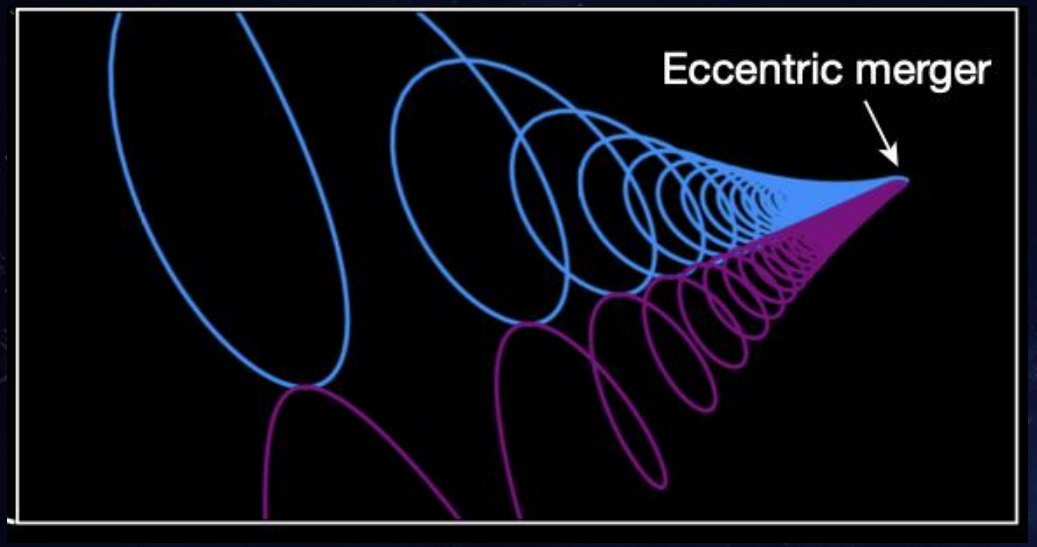


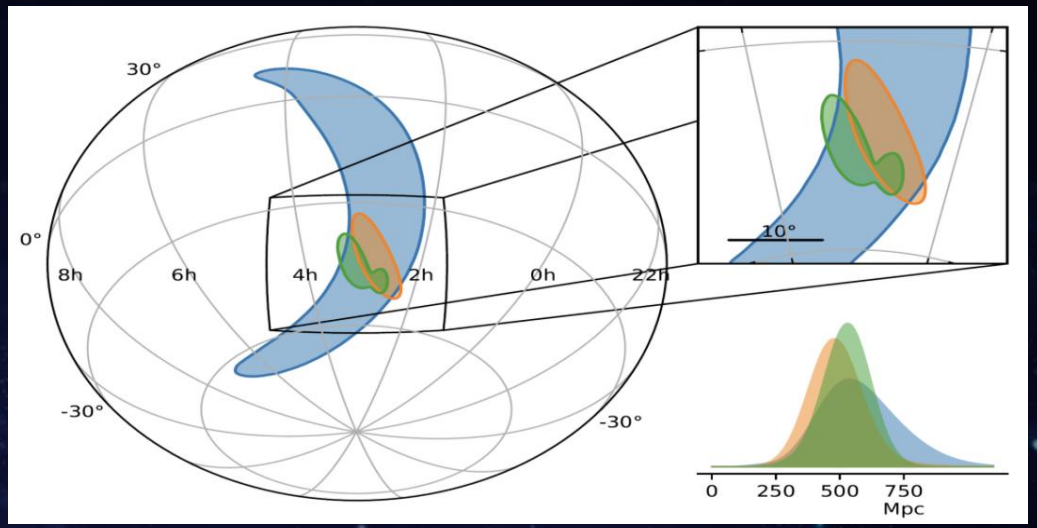
Gerosa & Fishbach 2021 Nature Astronomy

GW from Binary black hole mergers

- intrinsic: mass, spin magnitude, spin tilt angles
- orbit: eccentricity
- locations





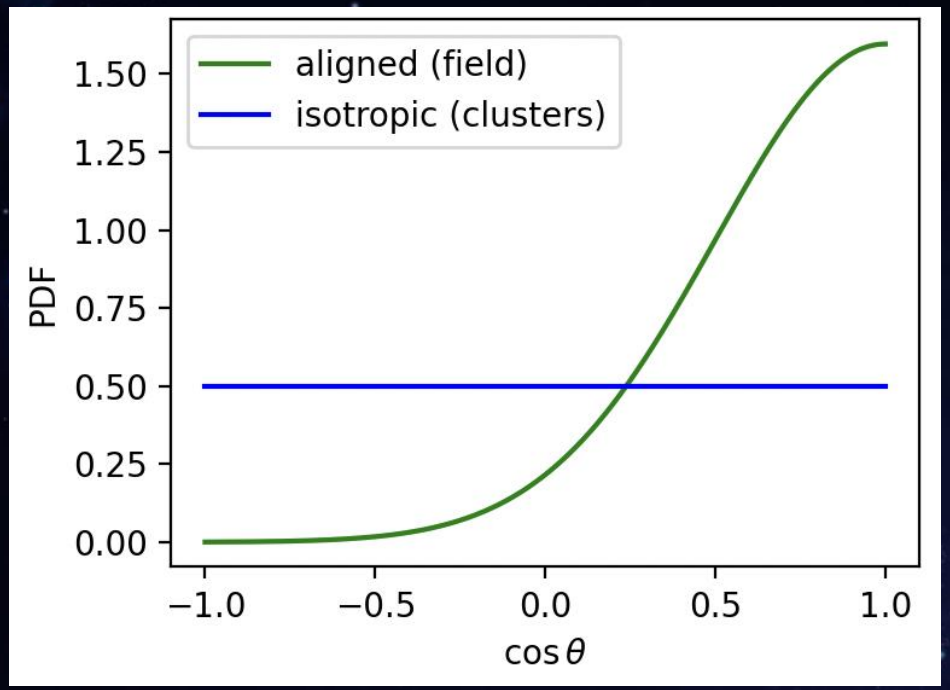


Outline

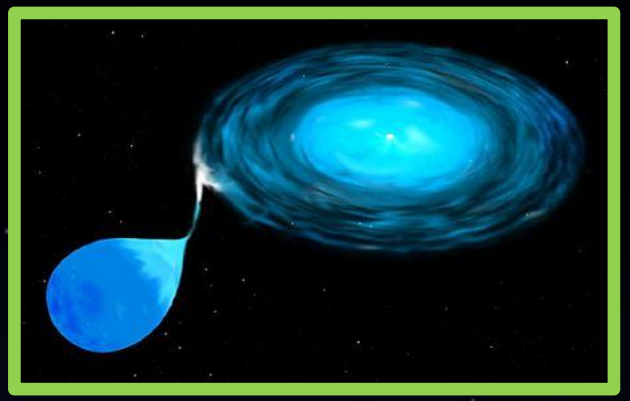
- Introduction：Formation channels of BBHs
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Signatures in gravitational waves

- signatures of spin orientations



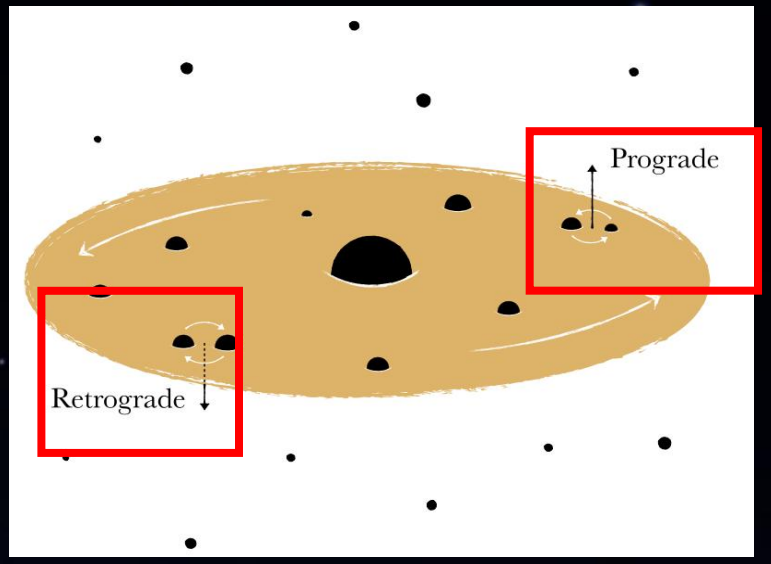
Rodriguez et al. 2016, ApJL

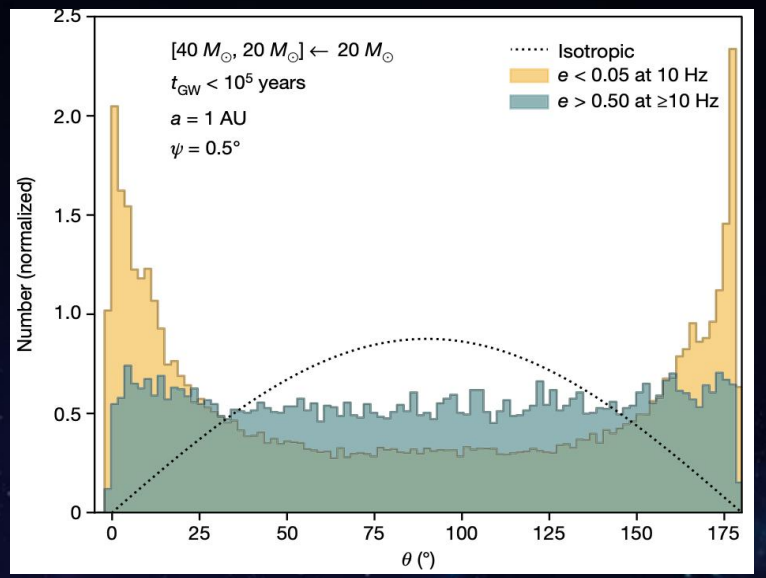


isolated field binaries



star clusters

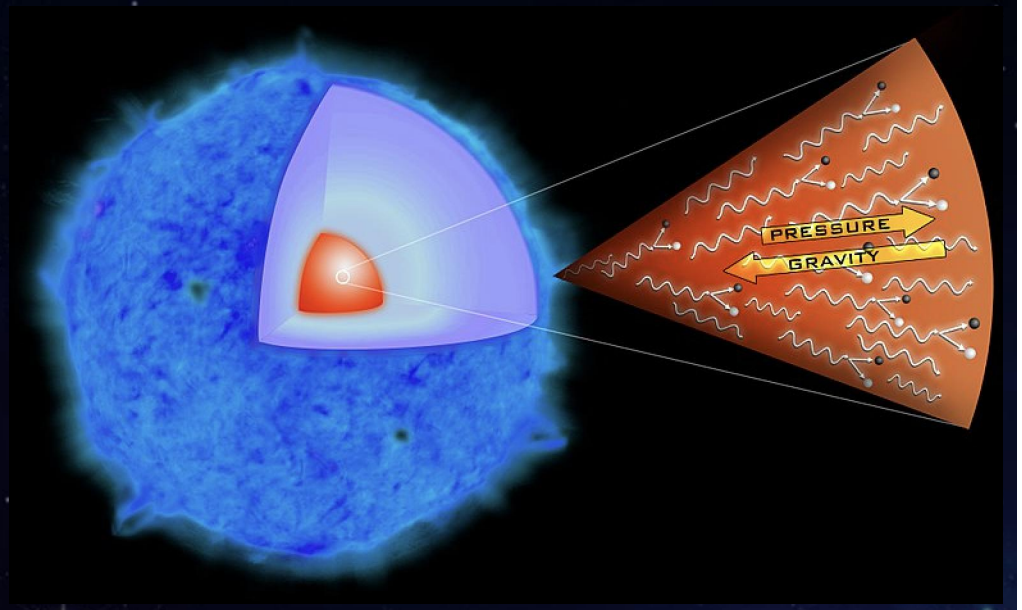




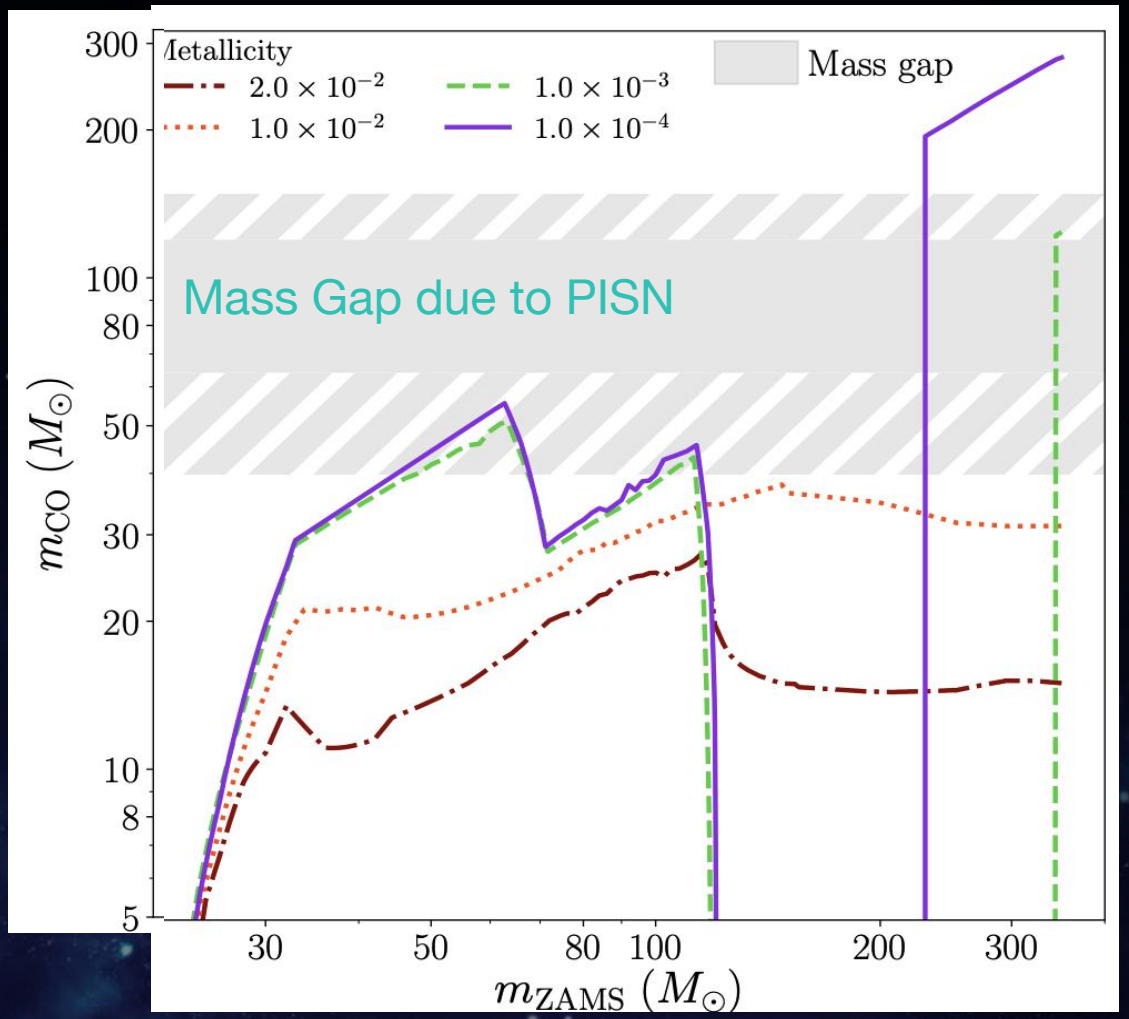
Samsing et al 2021, Nature

- signatures of mass distribution

Core-collapse BHs

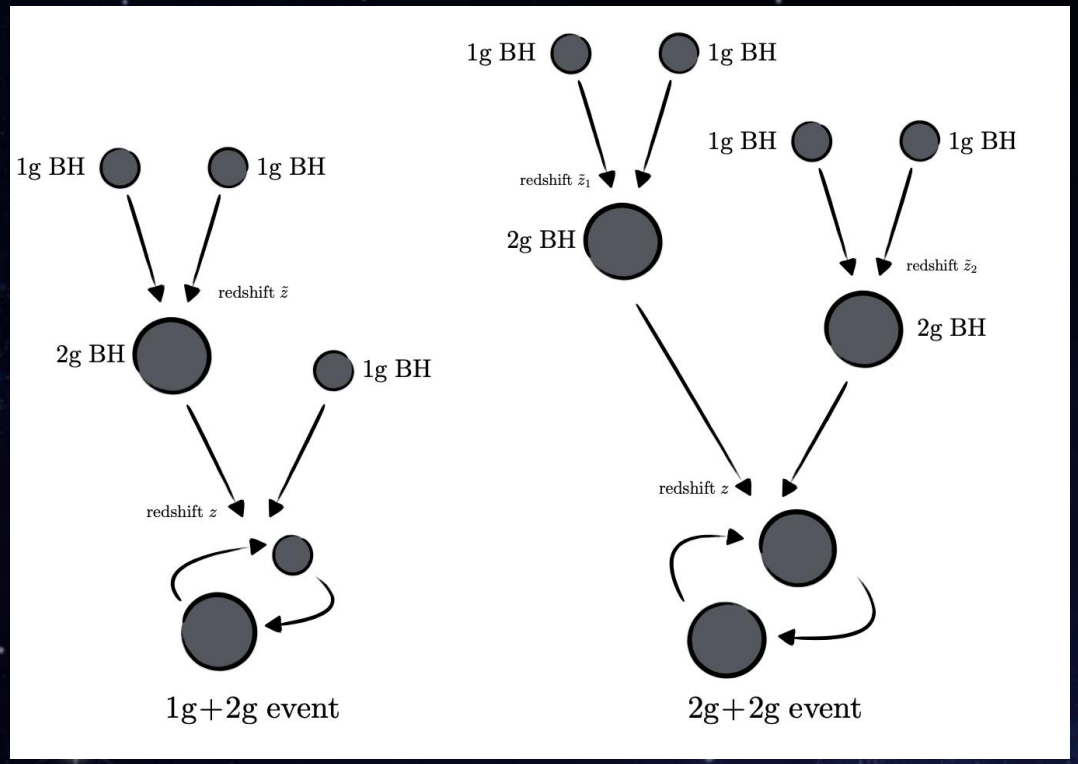


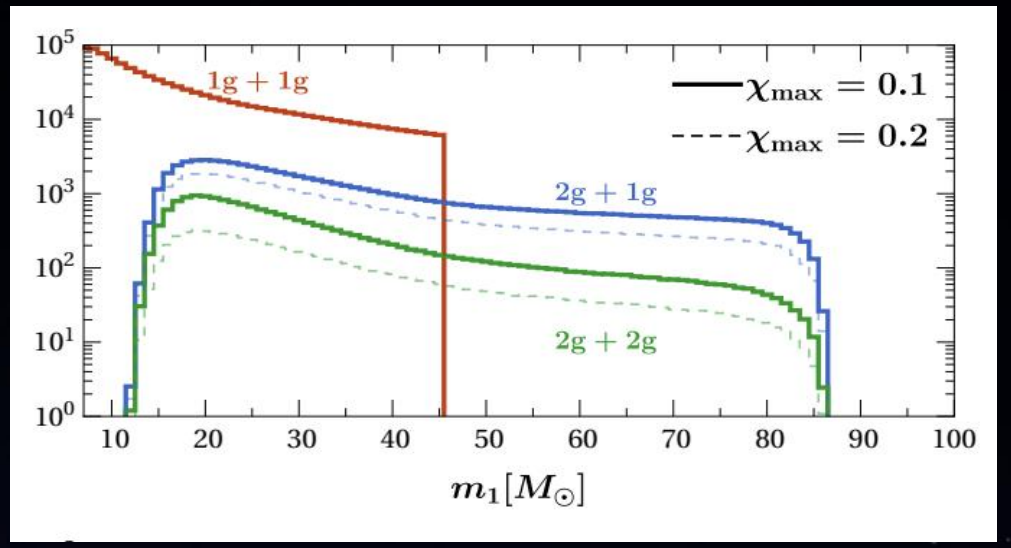
Pair-instability Supernovea (PISN)

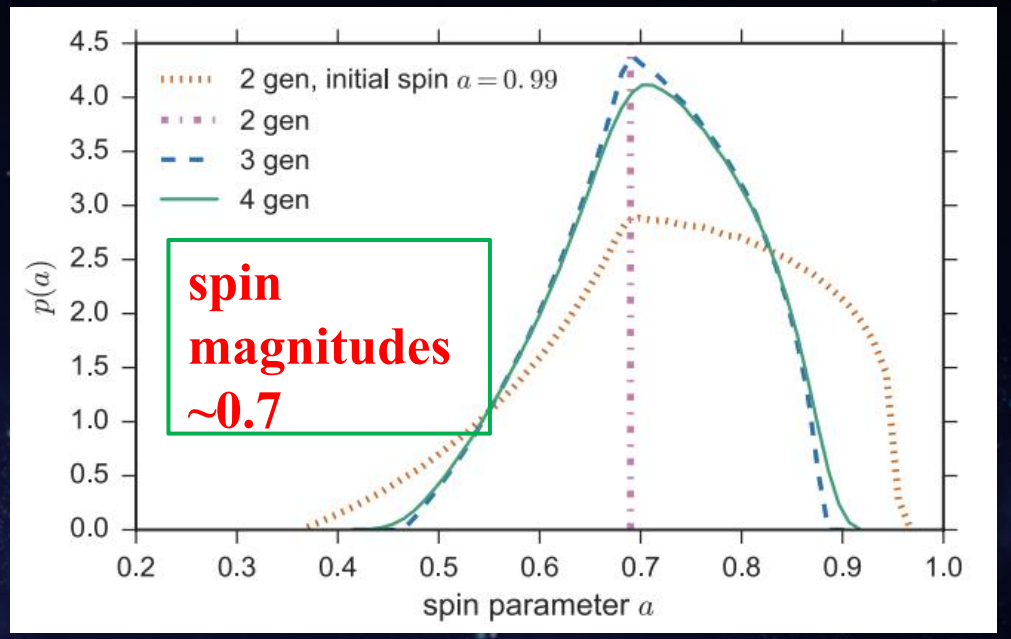


- signatures of mass and spin magnitude distribution

Repeated / Hierarchical mergers







Gerosa & Berti, 2017, PRD
Vishal et al., 2020, PRD
Fishbach, Holz & Farr 2017, ApJL

A brief summary:

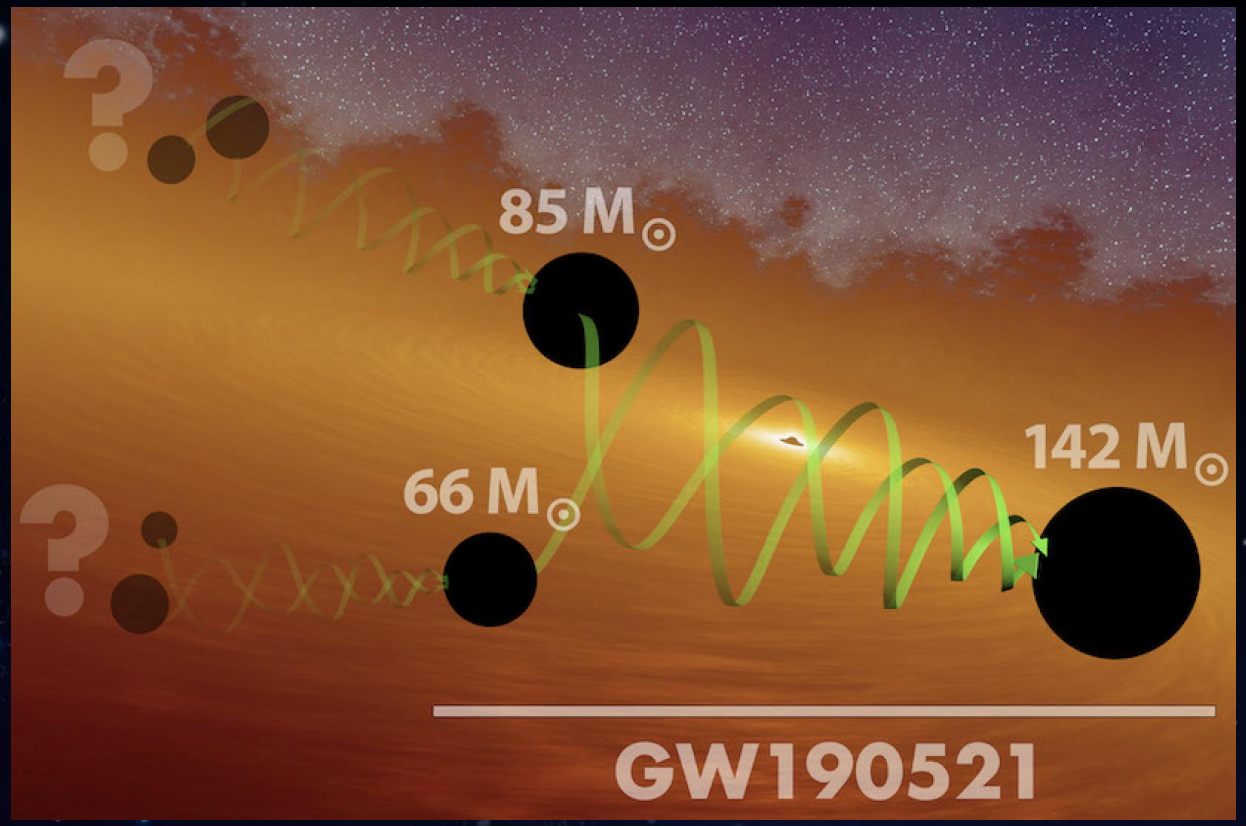
		Gas environmemnts \ Acceration?		
		Yes	No	
Deep Potential Well / High Density ?	Yes	AGN disk (gas-assist)	Star Clusters (dynamical)	Hierarchical mergers (high spin ~ 0.7)
	No	Isolated binary channels (Common evolope)		PISN mass gap (low spin)
		Aligned、Anti-aligned spins (larger spins?)	Isotropic spins	

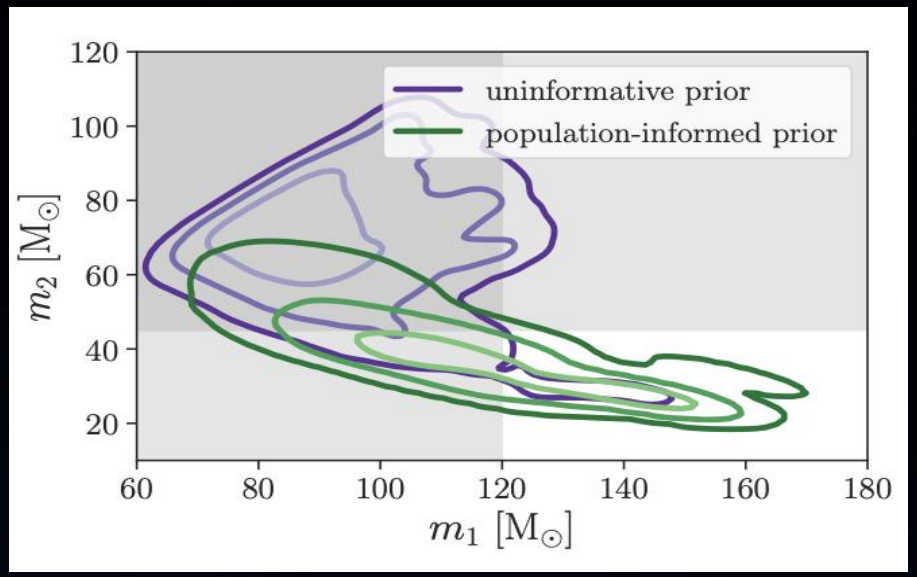
mass---spin magnitude---spin tilt

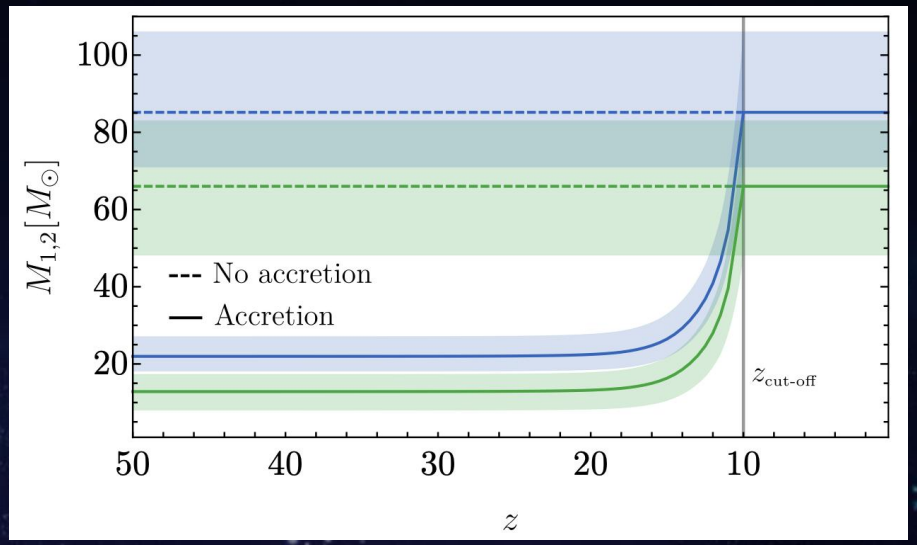
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GW190521： A straddling binary? hierarchcial merger? Acceration?





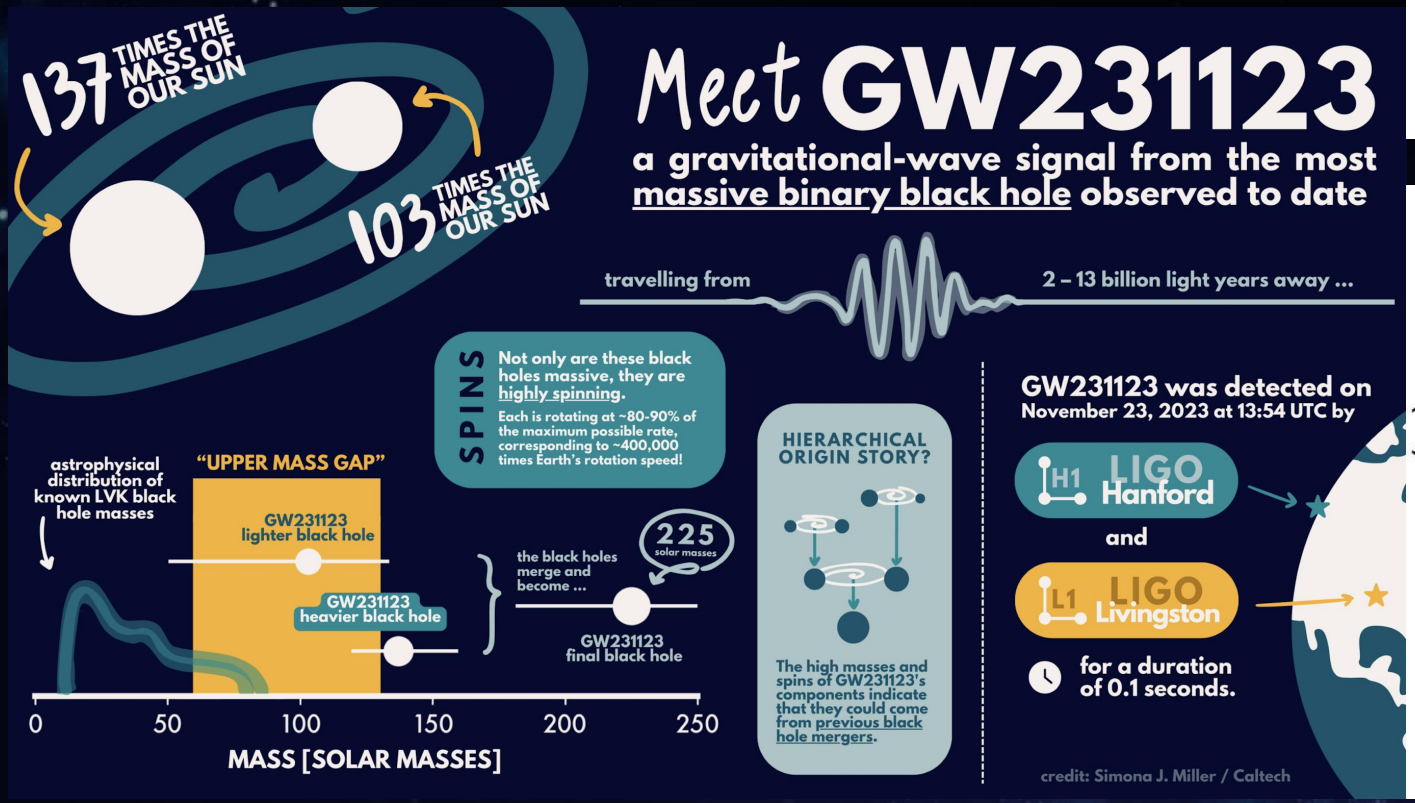


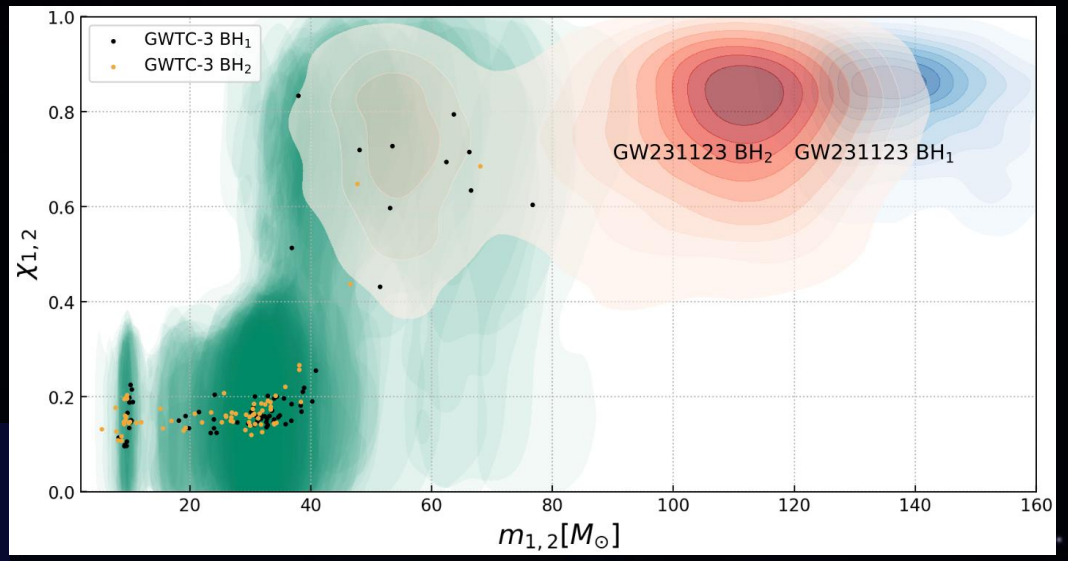
Abbott et al., 2020 ApJL

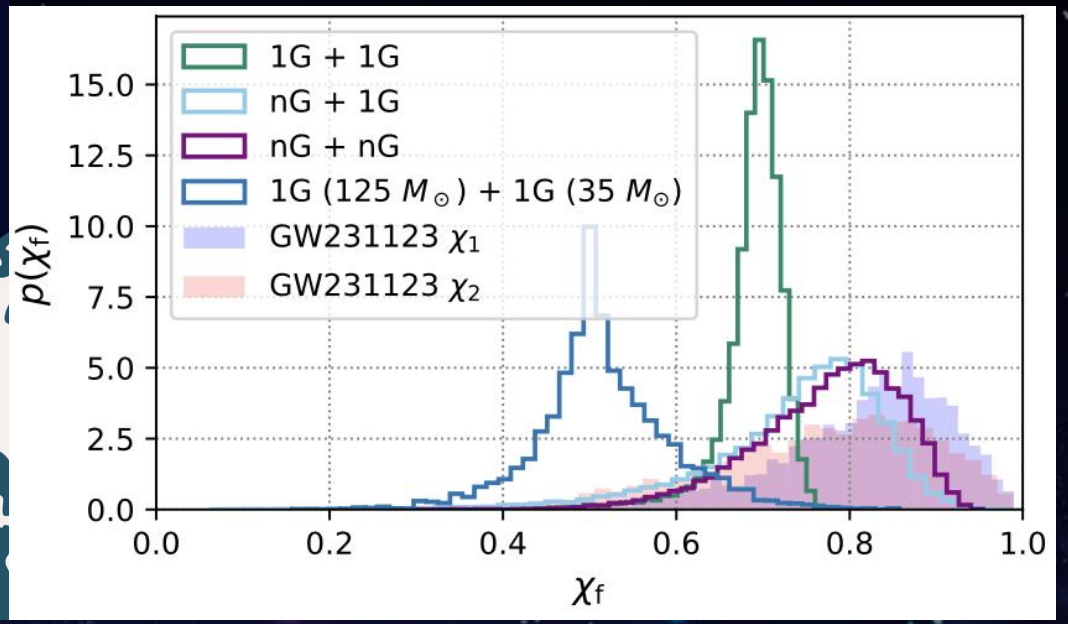
Fishbach&Daniel, 2020 ApJL

De Luca et al. 2021 PRL

GW231123: The LVK's most massive BBH : Hierarchical mergers ?





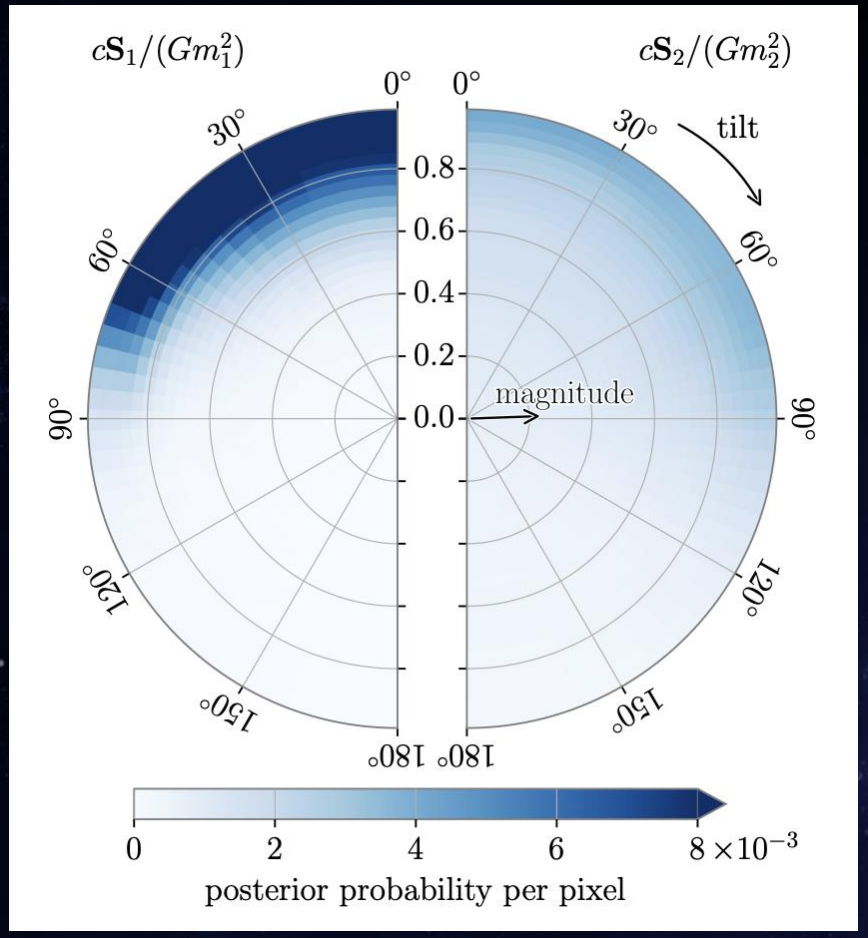


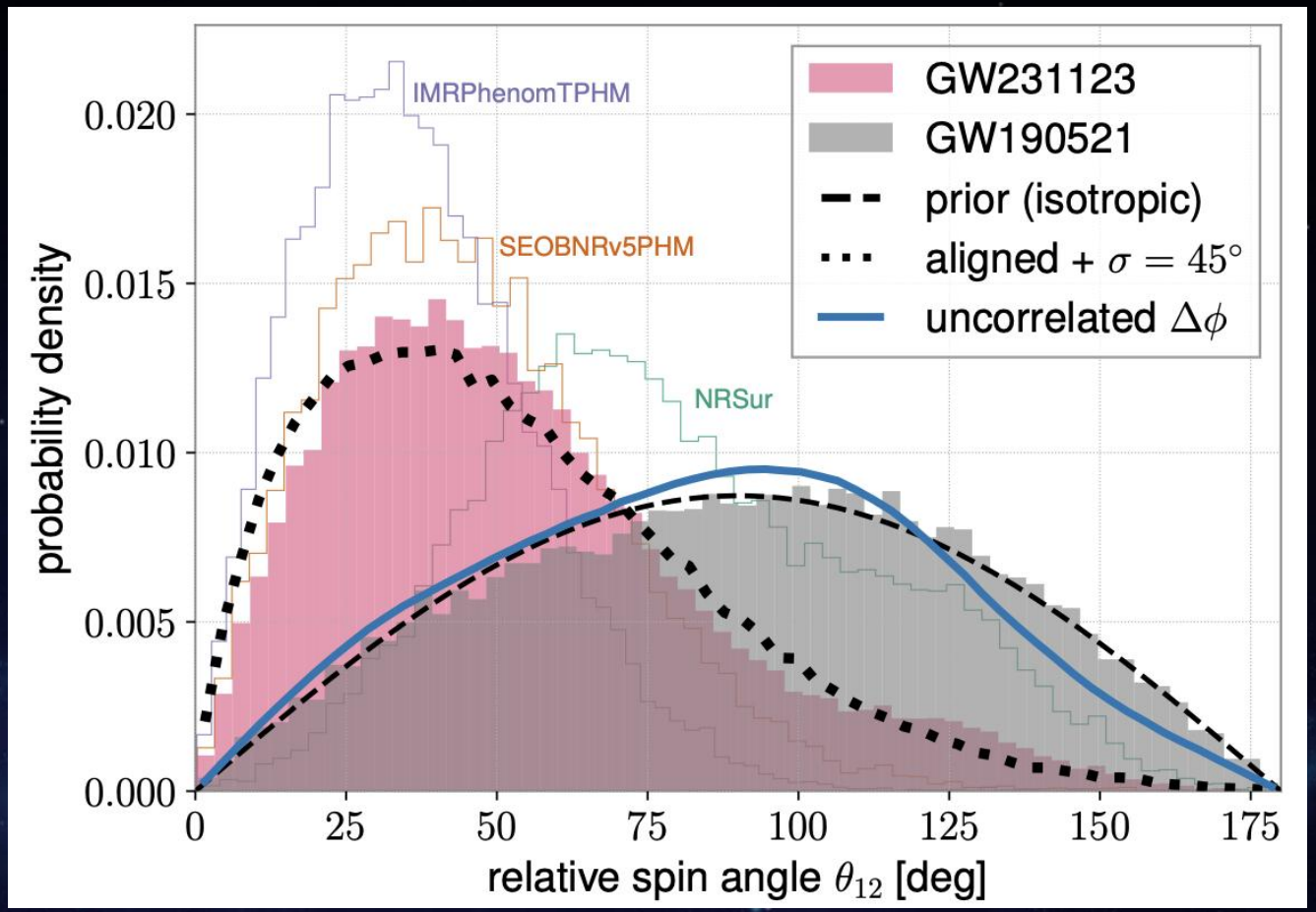
credit LVKC

Li et al. 2026 ApJ

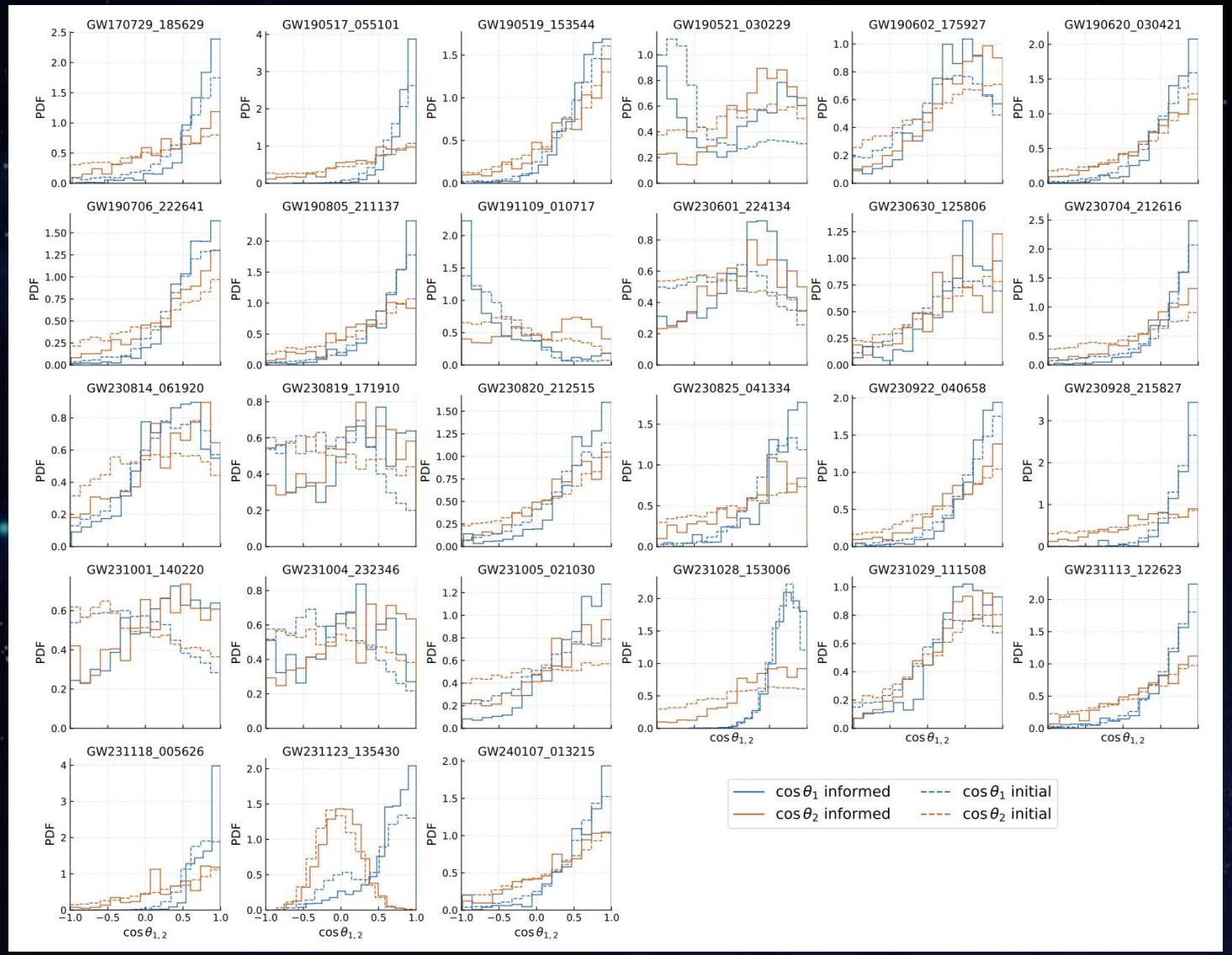
- 单个事件不能说明问题：

GW231123 来自aligned population？

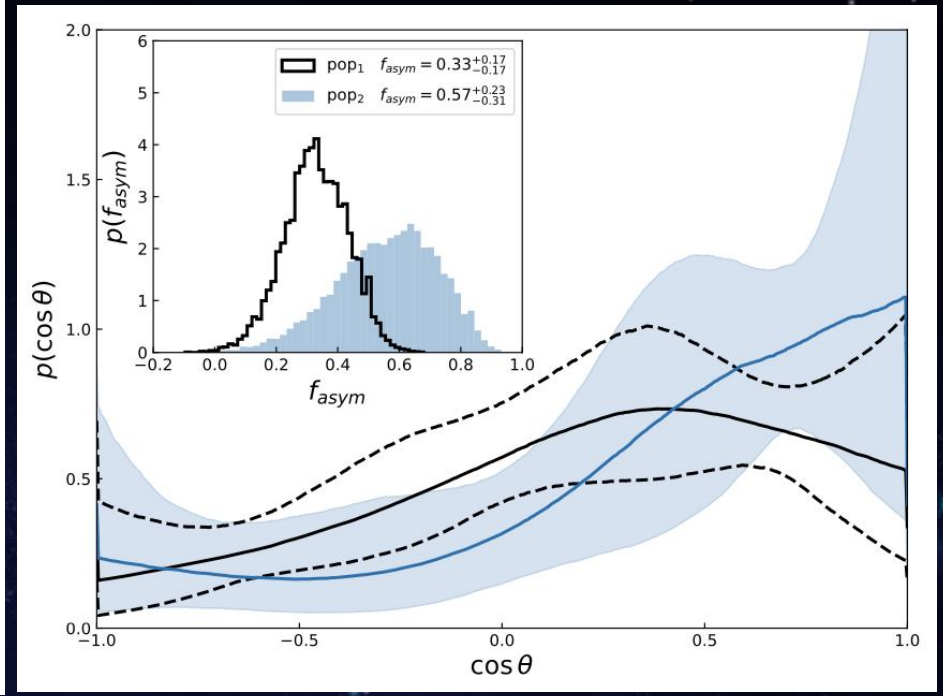




一群呢？如何得到它们的真实分布？



Hierarchical Bayesian Inference（分层贝叶斯推断）



我们为什么要用 分层贝叶斯推断？

- 事件测量误差比较大
- 单个事件不能说明问题

擅长：有测量误差的大样本分析

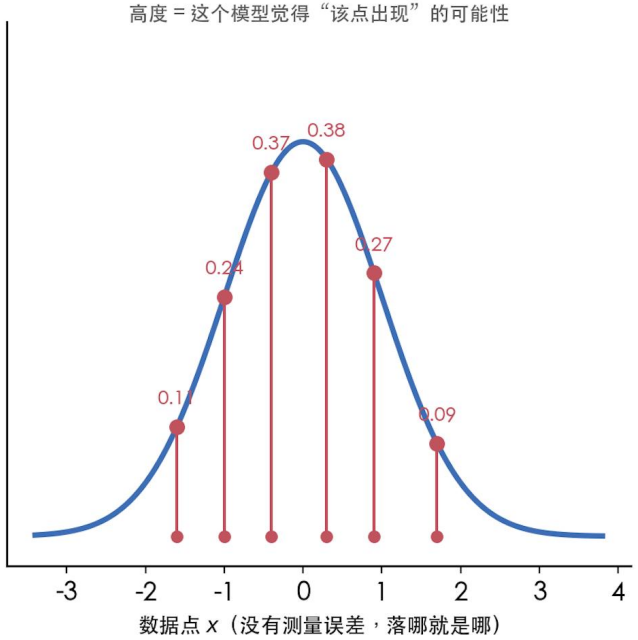
切入点：Likelihood，观测值（数据）没有误差情况

构造思路：随机抽样（相互独立）

似然：给每个模型打个分，看它“造出”这组数据有多容易

① 把每个点抬到曲线上，读高度

高度 = 这个模型觉得“该点出现”的可能性



② 把高度乘起来

$p(x_1) = 0.111$

$\times p(x_2) = 0.242$

$\times p(x_3) = 0.368$

$\times p(x_4) = 0.381$

$\times p(x_5) = 0.266$

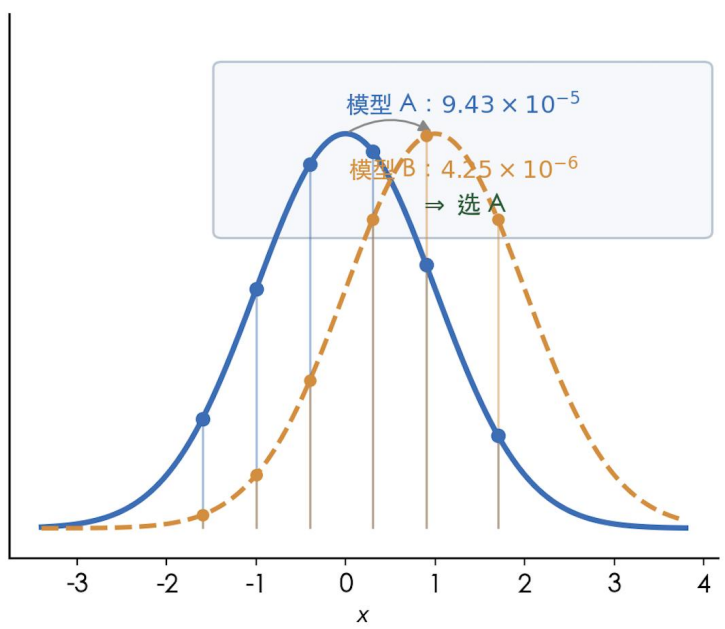
$\times p(x_6) = 0.094$

这个分数就是“模型 A 造出这组数据的可能性”

$= 似然 = 9.43 \times 10^{-5}$

模型 B：把曲线挪到右边

③ 换个模型再算一次，分数大者胜



$$L(\Lambda) = p(x_1 \mid \Lambda) \times p(x_2 \mid \Lambda) \times \cdots \times p(x_n \mid \Lambda) = \prod_{i=1}^{n} p(x_i \mid \Lambda) \qquad \ell(\Lambda) = \log L = \sum_{i=1}^{n} \log p(x_i \mid \Lambda)$$

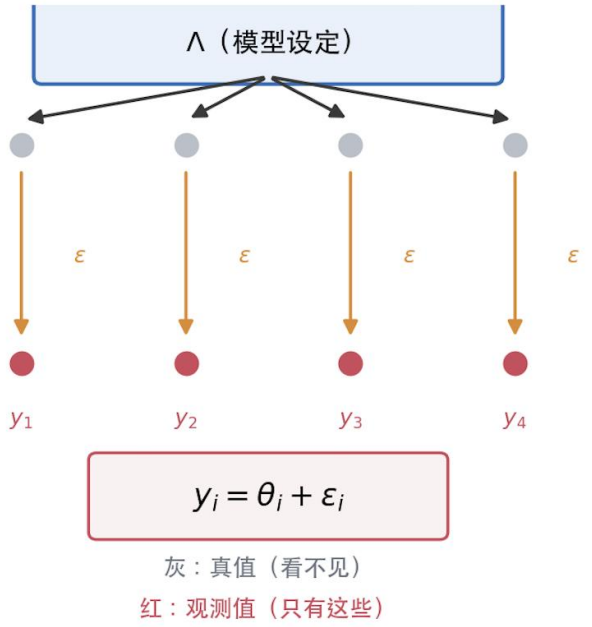
独立 → 相乘；一堆小于 1 的数相乘会变成 0，所以取对数：乘法变加法，答案不变，好算得多。

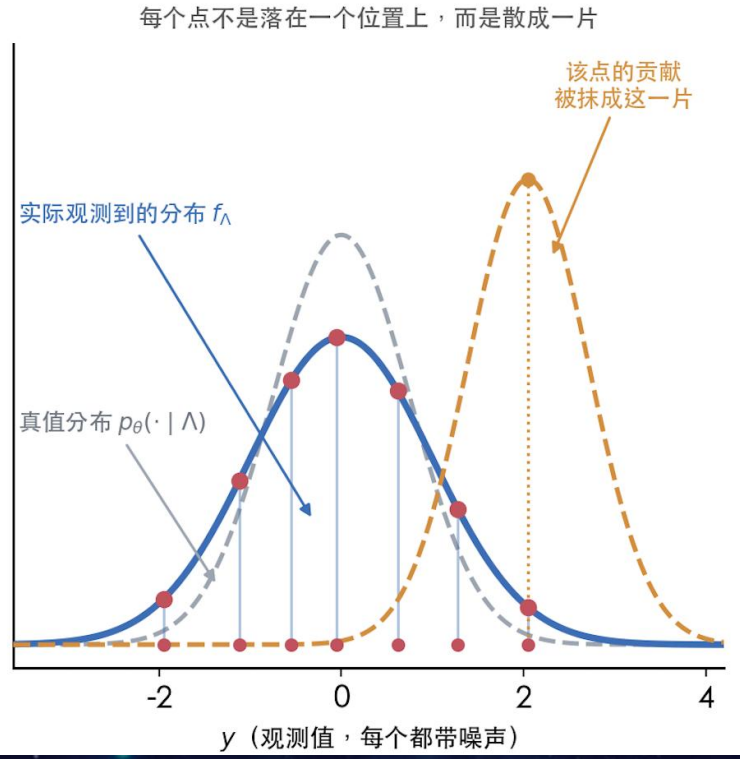
已知模型问数据 = 概率 | 已知数据问模型 = 似然 | 挑分数最高的模型 = 极大似然估计

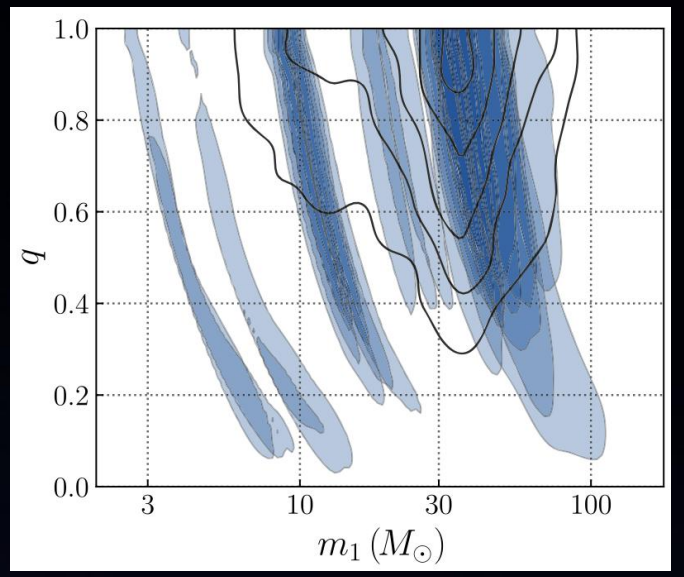
分层likelihodd的原因：观测值（数据）有误差

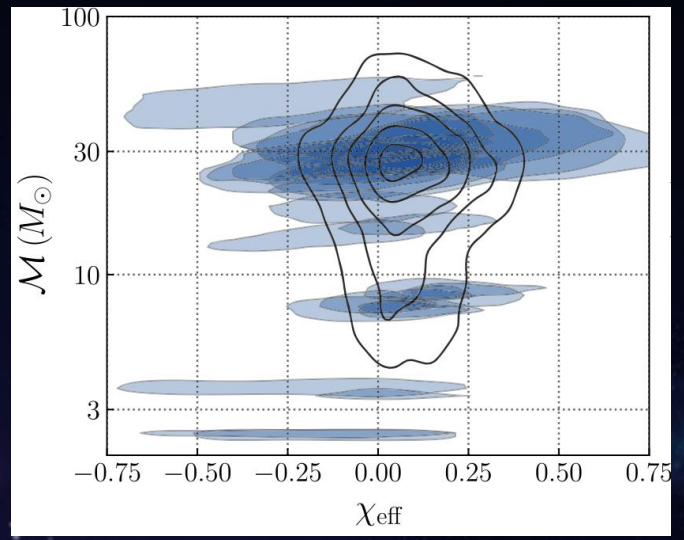
观测 = 真值 + 噪声：$y_i = \theta_i + \varepsilon_i$，其中真值 $\theta_i \sim p_\theta(\cdot \mid \Lambda)$、噪声 $\varepsilon_i \sim p_\varepsilon(\cdot \mid \sigma)$，$\theta_i$ **看不见。**

$$L(\Lambda, \sigma) = \prod_{i=1}^{n} \int \underbrace{p(y_i \mid \theta_i, \sigma)}_{\text{噪声层}} \underbrace{p(\theta_i \mid \Lambda)}_{\text{真值层}} d\theta_i = \prod_{i=1}^{n} f_\Lambda(y_i)$$

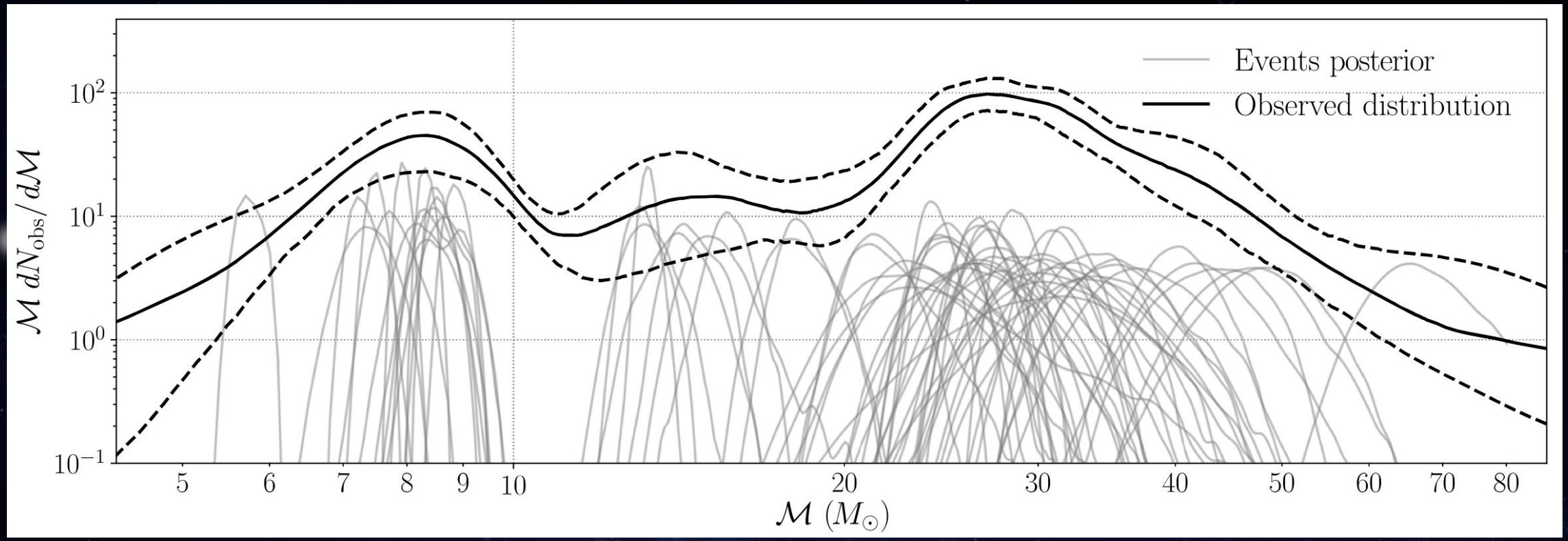








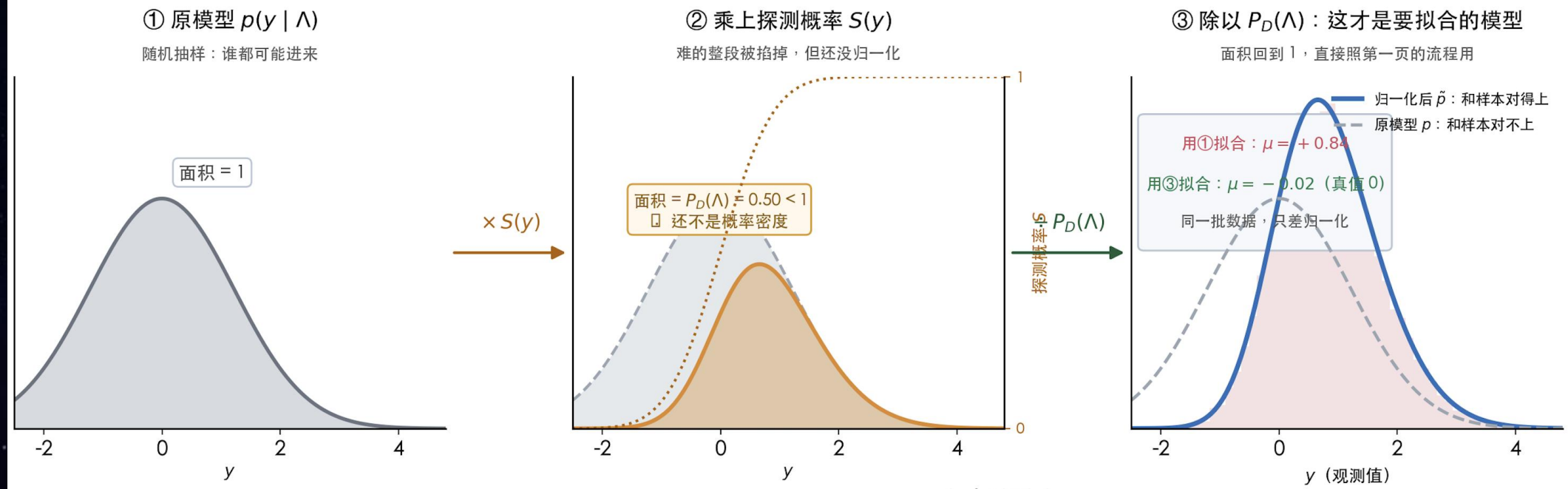
直观感受：推断结果



LVKC 2023 PRX

考虑选择效应：重的黑洞更容易被探测到

选择效应：把「原始模型」换成「能被探测到的模型」，再归一化

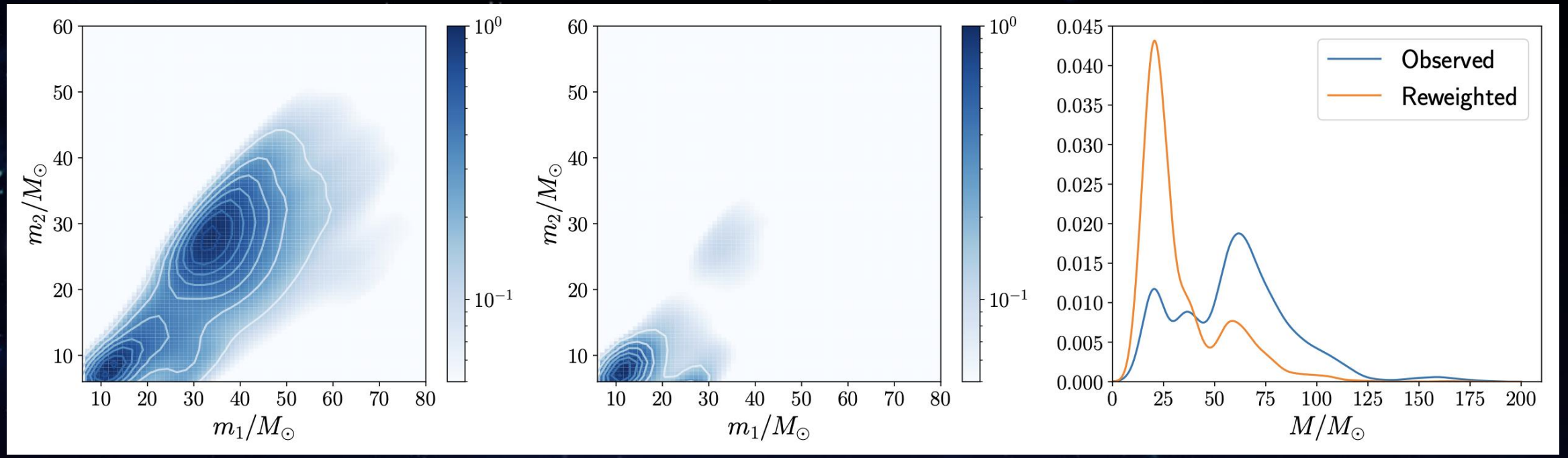


原始模型 $p(y \mid \Lambda)$ → 乘上探测概率 $S(y)$ → 除以 $P_D(\Lambda)$ → 得到 $\tilde{p}(y \mid \Lambda) = \dfrac{p(y \mid \Lambda) S(y)}{P_D(\Lambda)}$，其中 $P_D(\Lambda) = \int p(y \mid \Lambda) S(y)\, dy$

$$L(\Lambda) = \prod_{i=1}^{n} \tilde{p}(y_i \mid \Lambda) \quad \Longleftrightarrow \quad \ell(\Lambda) = \sum_i \log p(y_i \mid \Lambda) + \sum_i \log S(y_i) - n \log P_D(\Lambda)$$

只有最后一步带 Λ，丢不掉；S 必须已知——否则模型只取决于乘积 pS，怎么配都行，数据分不开

直观感受：选择效应



Nitz et al. 2021 ApJ

Thanks!

一句话讲透

似然函数 = 给每个候选模型打的一个"分数"，分数越高说明这个模型越容易"造出"你手上这组数据。

换个更好懂的比喻：模型是一台**造数据的机器**。你有 n 个数据点，就问这台机器——"你造出这组点的可能性有多大？"这个可能性就是似然。

概率 vs 似然（初学者最容易绕晕的地方）：

- 概率：模型已知，问"数据长什么样"
- 似然：数据已知，问"该选哪个模型"

为什么是"乘"起来的?

因为**独立事件同时发生的概率 = 各自概率相乘**。

抛两次硬币都是正面：$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$。抽 n 个样本是 n 件独立的事，所以把它们各自的可能性乘起来：

$$L(\Lambda) \;=\; p(x_1 \mid \Lambda) \times p(x_2 \mid \Lambda) \times \cdots \times p(x_n \mid \Lambda) \;=\; \prod_{i=1}^{n} p(x_i \mid \Lambda)$$

这里的 Λ 就是"模型"，可以是一个数（比如均值），也可以是一整套设定（分布形状、参数……）。

"没有测量误差"帮了什么忙? 数据点就是真值，你不用先猜"它本来在哪、偏了多少"。点上哪儿，就直接抬到曲线上读数——中间没有任何环节。有测量误差的话，每一步都得先"糊"一下再读，对初学者可以先不管。

从“单个事件”到“总体分布”：

单事件参数估计：

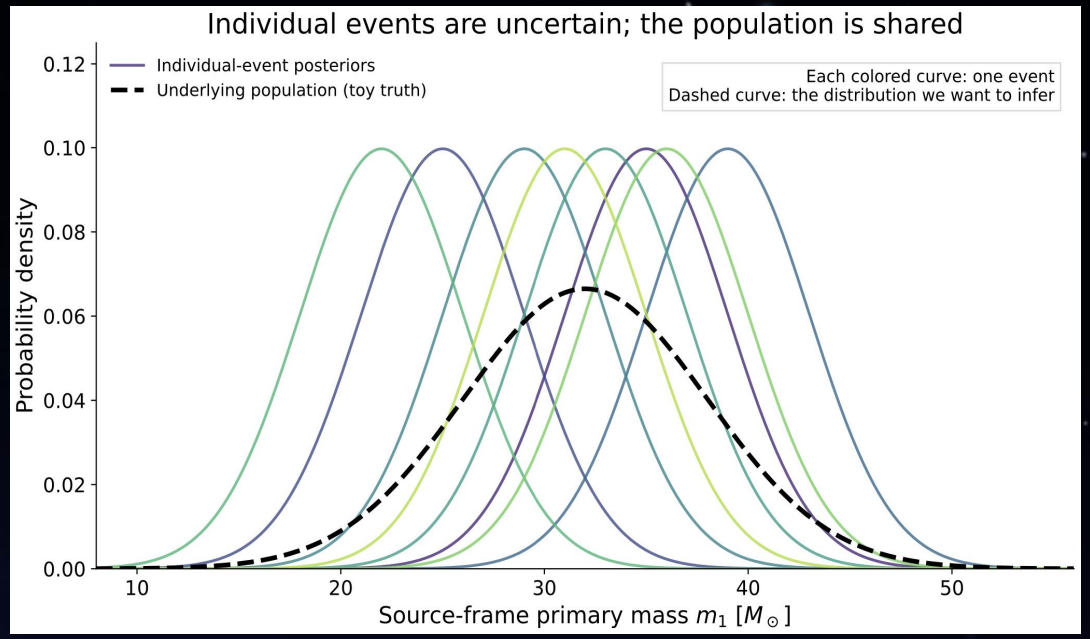
$$p(\theta_i \mid d_i) = \frac{p(d_i \mid \theta_i)\,\pi_{\mathrm{PE}}(\theta_i)}{Z_i}$$

总体模型：

$$\theta_i \sim p_{\mathrm{pop}}(\theta \mid \Lambda)$$

其中：

- d_i：第 i 个事件的数据；
- θ_i：该事件的参数，如质量、自旋、红移；
- π_{PE}：单事件参数估计使用的先验；
- Λ：描述总体分布的**超参数**，例如质量分布斜率、峰的位置和宽度。



- 关键困难：每个事件的参数都有测量不确定性。

-------- 分层贝叶斯分析

分层贝叶斯方法:

单个事件误差+群体分布

总体超参数 Λ $\longrightarrow$ 事件真实参数 θ_i $\longrightarrow$ 观测数据 d_i

单个事件对总体模型的支持程度:

$$\mathcal{L}_i(\Lambda) = \int p(d_i \mid \theta_i)\, p_{\rm pop}(\theta_i \mid \Lambda)\, \mathrm{d}\theta_i$$

多个独立事件联合约束总体:

$$p(\Lambda \mid d_1, \ldots, d_N) \propto \pi(\Lambda) \prod_{i=1}^{N} \mathcal{L}_i(\Lambda)$$

若单事件后验样本为 $\theta_i^{(k)}$，则:

$$\mathcal{L}_i(\Lambda) \approx \frac{Z_i}{K_i} \sum_{k=1}^{K_i} \frac{p_{\rm pop}\left(\theta_i^{(k)} \mid \Lambda\right)}{\pi_{\rm PE}\left(\theta_i^{(k)}\right)}$$

考虑选择效应

总体模型对应的探测效率:

$$\alpha(\Lambda) = \int P_{\rm det}(\theta)\, p_{\rm pop}(\theta \mid \Lambda)\, \mathrm{d}\theta$$

被探测事件的真实参数分布:

$$p(\theta \mid \mathrm{det}, \Lambda) = \frac{P_{\rm det}(\theta)\, p_{\rm pop}(\theta \mid \Lambda)}{\alpha(\Lambda)}$$

条件于已探测事件数 N、只推断总体分布形状时:

$$p(\Lambda \mid \{d_i\}, \mathrm{det}, N) \propto \pi(\Lambda) \frac{\prod_{i=1}^{N} \int p(d_i \mid \theta_i)\, p_{\rm pop}(\theta_i \mid$$
$$\alpha(\Lambda)^N$$

Thrane & Talbot, 2019, PASA （1809.02293）

探测到的总体，不等于真实总体：必须考虑“哪些事件更容易被发现”

总体模型对应的探测效率：

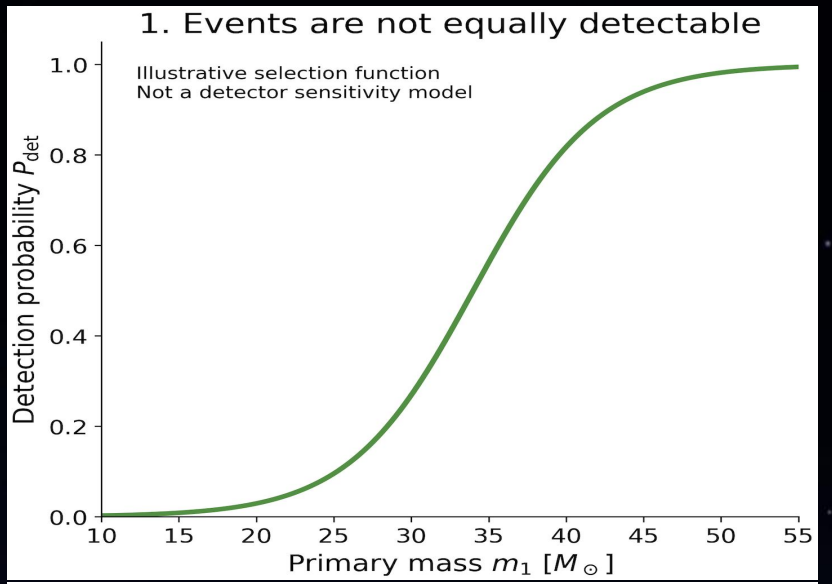
$$\alpha(\Lambda) = \int P_{\rm det}(\theta)\, p_{\rm pop}(\theta \mid \Lambda)\, {\rm d}\theta$$

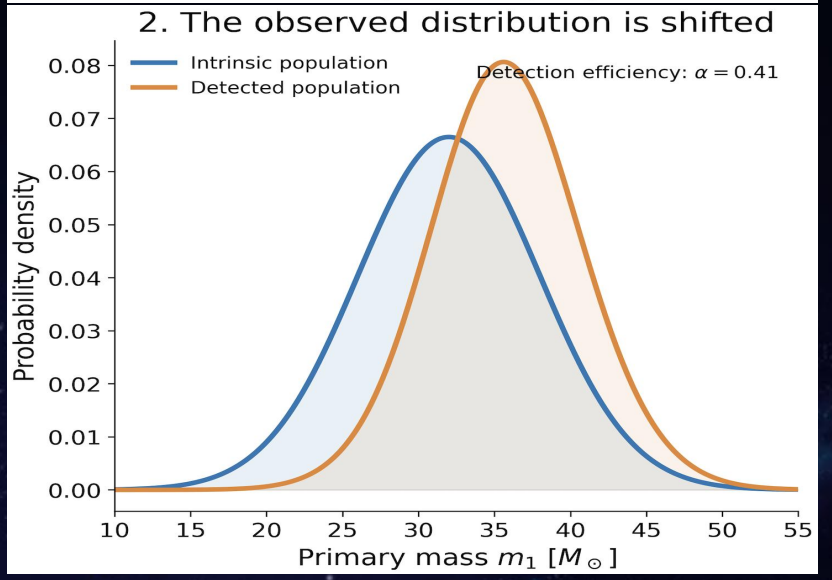
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分层贝叶斯方法:

无选择效应

总体超参数 Λ $\longrightarrow$ 事件真实参数 θ_i $\longrightarrow$ 观测数据 d_i

单个事件对总体模型的支持程度:

$$\mathcal{L}_i(\Lambda) = \int p(d_i \mid \theta_i)\, p_{\rm pop}(\theta_i \mid \Lambda)\, \mathrm{d}\theta_i$$

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考虑选择效应

总体模型对应的探测效率:

$$\alpha(\Lambda) = \int P_{\rm det}(\theta)\, p_{\rm pop}(\theta \mid \Lambda)\, \mathrm{d}\theta$$

被探测事件的真实参数分布:

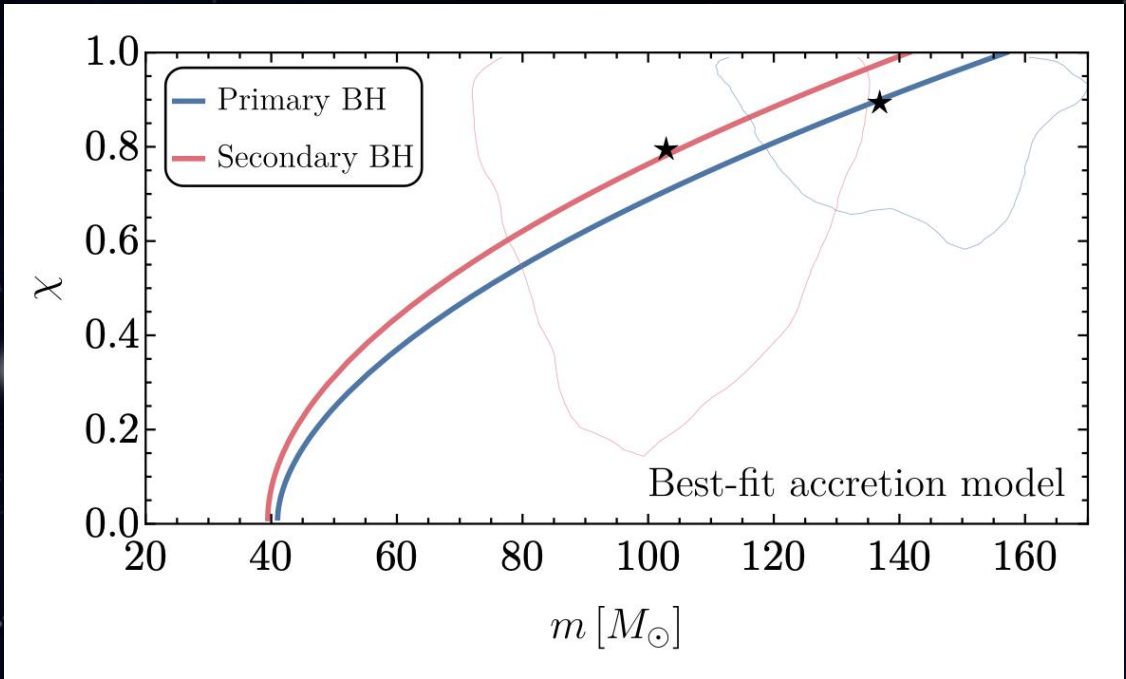
$$p(\theta \mid \mathrm{det}, \Lambda) = \frac{P_{\rm det}(\theta)\, p_{\rm pop}(\theta \mid \Lambda)}{\alpha(\Lambda)}$$

条件于已探测事件数 N、只推断总体分布形状时:

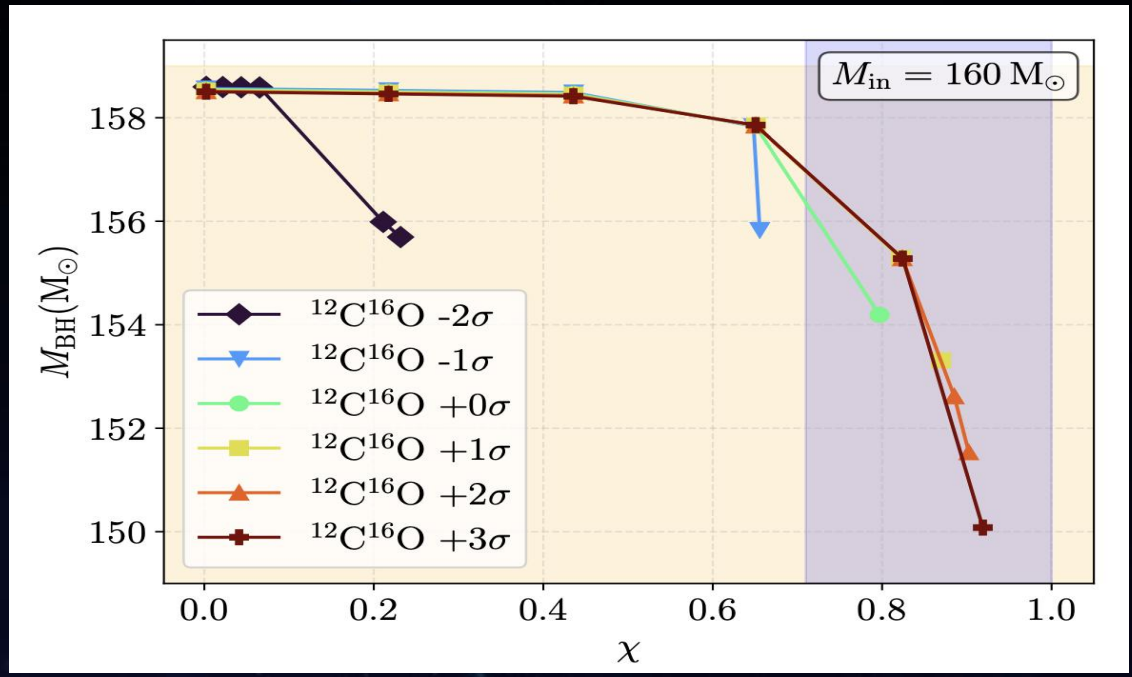
$$p(\Lambda \mid \{d_i\}, \mathrm{det}, N) \propto \pi(\Lambda) \frac{\prod_{i=1}^{N} \int p(d_i \mid \theta_i)\, p_{\rm pop}(\theta_i \mid \Lambda)\, \mathrm{d}\theta_i}{\alpha(\Lambda)^N}$$

Thrane & Talbot, 2019, PASA

Various formation channels are possible :
PBH+accretion \ Rotating star formation



Luca et al. 2026 PRL



Croon et al. 2026 MNRAS